24EECE2001 – Electronic Devices and Circuits

BIPOLAR JUNCTION TRANSISTORS

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Source:
Adel S. Sedra, Kenneth C. Smith, Arun N. Chandorkar, Microelectronic Circuits, 7/e, Oxford University Press, 2017

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Syllabus

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MODULE 2 DIODES AND DIODE CIRCUITS The Ideal Diode, Terminal Characteristics of Junction Diodes, Modeling the Diode Forward Characteristic, Operation in the Reverse Breakdown Region—Zener Diodes, Rectifier Circuits, Limiting and Clamping Circuits

MODULE 3 BIPOLAR JUNCTION TRANSISTORS Device Structure and Physical Operation, Current–Voltage Characteristics, BJT Circuits at DC, Applying the BJT in Amplifier Design, Small-Signal Operation and Models, Basic BJT Amplifier Configurations. Transistor Breakdown and temperature effects

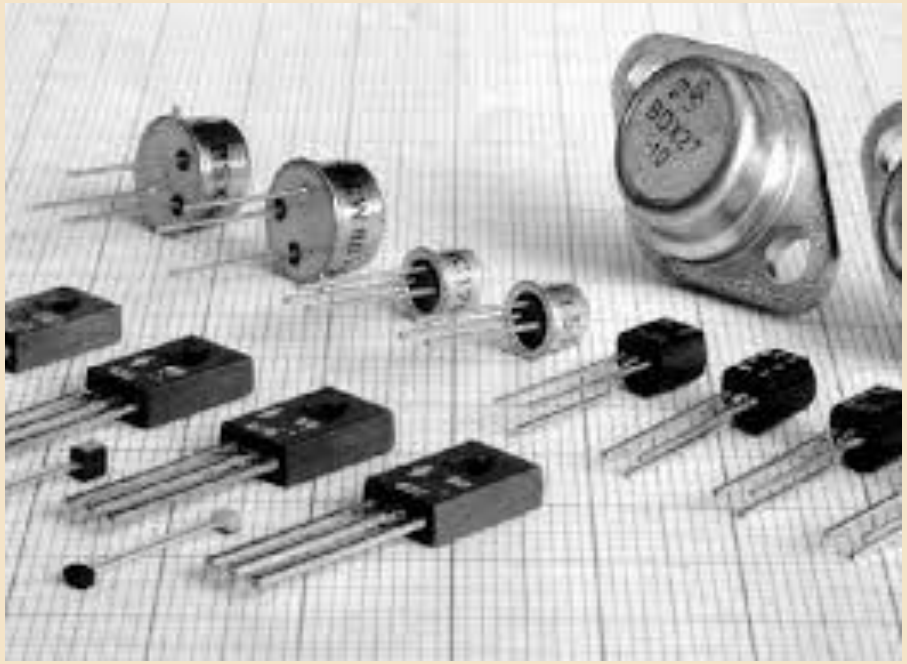
MODULE 4 MOS FIELD EFFECT TRANSISTORS Device Structure and Physical Operation, Current—Voltage Characteristics, MOSFET Circuits at DC, Applying the MOSFET in Amplifier Design, Small-Signal Operation and Models, Basic MOSFET Amplifier Configurations, Biasing in MOS Amplifier Circuits, Discrete-Circuit MOS Amplifiers

MODULE 5 AMPLIFIER FREQUENCY RESPONSE Review of Transfer function, Frequency Response, Pole-Zero Concepts, Transfer function and AC Analysis of a First Order RC Circuit, Capacitive Effects in a PN Junction, Internal Capacitive Effects and the High-Frequency Model of the MOSFET, High-Frequency Response of the Common Source Amplifiers, High Frequency Response of Common Gate Amplifiers, High Frequency Response of Source Followers

Introduction



Vacuum Tubes



Transistor Packages

Vacuum tube = large + hot + power hungry
A considerable amount of power is wasted as heat

The transistor was invented at Bell Laboratories in 1947 by John Bardeen, Walter Brattain, and William Shockley.
MOSFET was familiar in 1970s and 1980s.

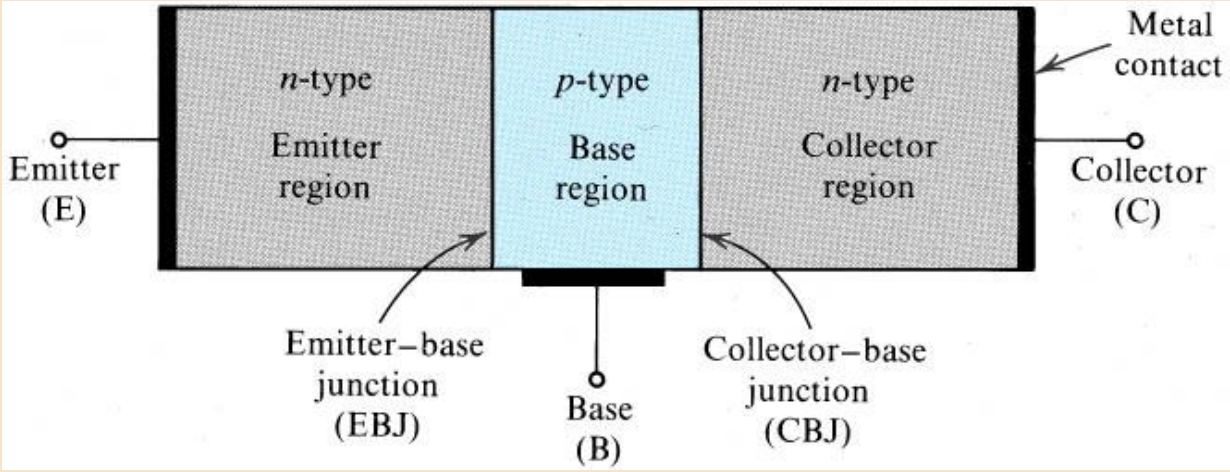
By 2014, the MOSFET was undoubtedly the most widely used electronic device, and CMOS technology the technology of choice in the design of integrated circuits.

BJT is still the preferred device in some very demanding analog and digital integrated-circuit applications. This is especially true in very-high-frequency and high-speed circuits.

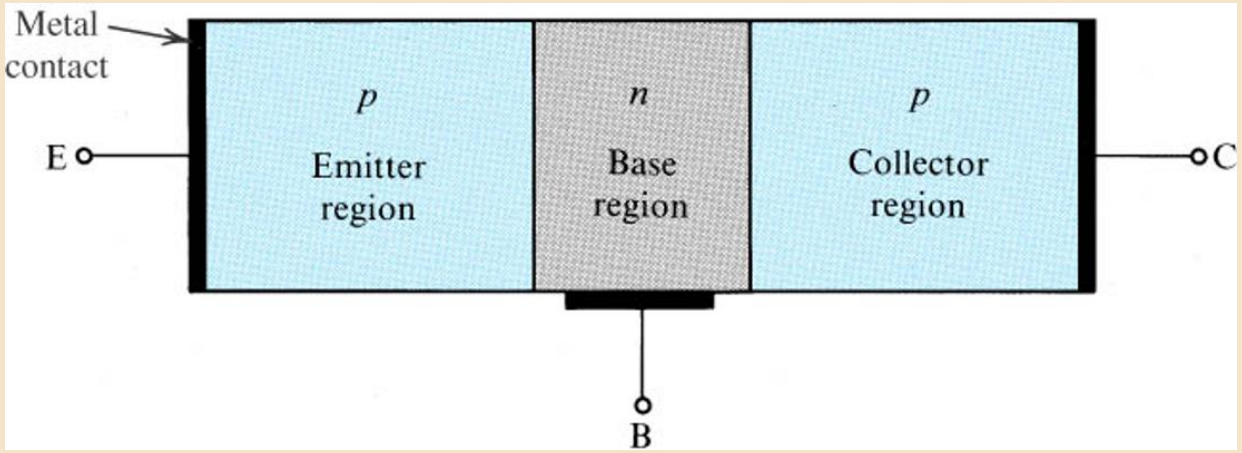
Finally, bipolar transistors can be combined with MOSFETs to create innovative circuits that take advantage of the high-input-impedance and low-power operation of MOSFETs and the very-high-frequency operation and high-current-driving capability of bipolar transistors. The resulting technology is known as BiCMOS, and it is finding increasingly larger areas of application.

BJT + CMOS = BiCMOS

DEVICE STRUCTURE AND PHYSICAL OPERATION

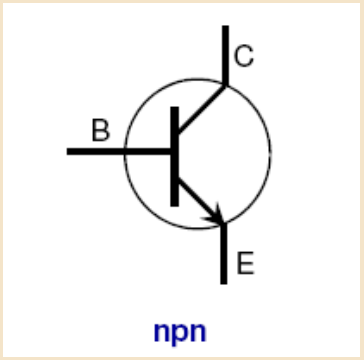


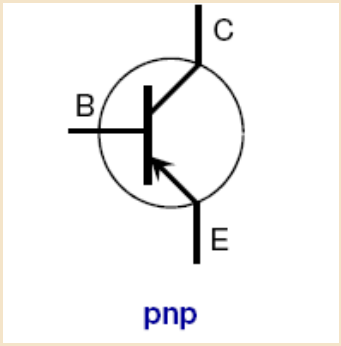
A simplified structure of the npn transistor
Emitter is heavily doped and base is lightly doped



A simplified structure of the pnp transistor

As charge carriers of both polarities-that is, electrons and holes-participate in the current-conduction process in a transistor, it is named as bipolar transistor.





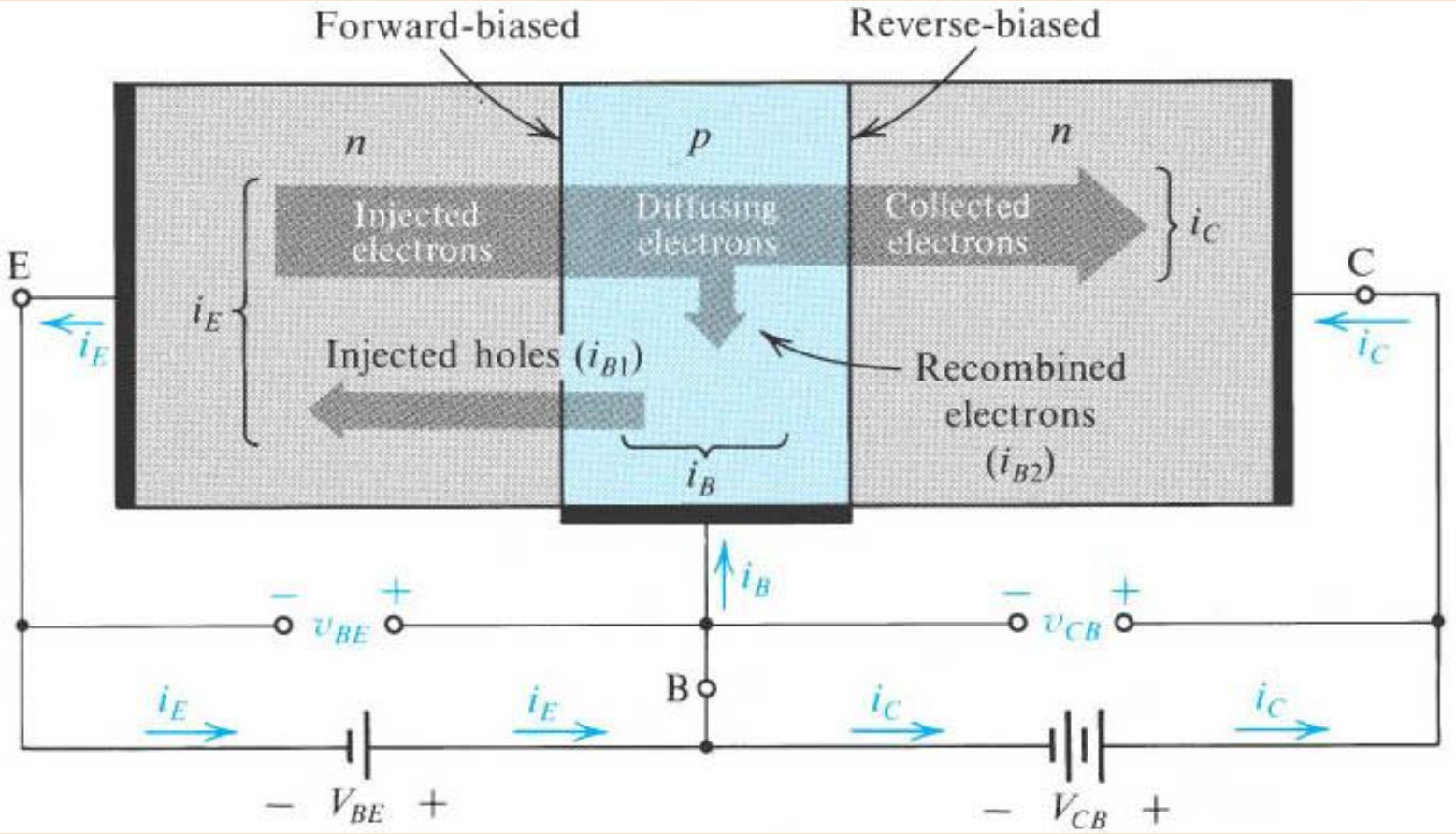
Modes of operation

Depending on the bias condition (forward or reverse) of each of the junctions, BJT operates in different modes.

Mode	Emitter Base Junction	Collector Base Junction
Active	Forward	Reverse
Saturation	Forward	Forward
Cut-off	Reverse	Reverse

- *The active mode used if the transistor is to operate as an amplifier.*
- *Both the cutoff and the saturation modes are used for Switching applications (e.g., logic circuits).*

Operation of the NPN transistor in the Active Mode



Current flow in an npn transistor biased to operate in the active mode

Emitter is highly doped than the base region i.e. the device is designed to have a high density of electrons in the emitter and a low density of holes in the base.

The Collector Curre $i_C = I_S e^{v_{BE}/V_T}$

The Base Current The base current i_B is composed of two components. The first component i_{B1} is due to the holes injected from the base region into the emitter region. This current component is proportional to e^{v_{BE}/V_T}

The second component of base current, i_{B2} , is due to holes that have to be supplied by the external circuit in order to replace the holes lost from the base through the recombination process. Because i_{B2} is proportional to the number of electrons injected into the base, it also will be proportional to e^{v_{BE}/V_T}

Thus the total base current, $i_B = i_{B1} + i_{B2}$, will be proportional to, e^{v_{BE}/V_T} and can be expressed as a fraction of the collector current i_C as follows:

$$i_B = \frac{i_C}{\beta} = i_B = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

where β is a transistor parameter, called the common-emitter current gain. For modern npn transistors, β is in the range 50 to 200, but it can be as high as 1000 for special devices.

The value of β is highly influenced by two factors:

- The width of the base region, W

- The relative dopings of the base region and the emitter region, N_A /N_D.

To obtain a high β (which is highly desirable since β represents a gain parameter) the base should be thin (W small) and lightly doped and the emitter heavily doped (making N_A /N_D small).

For modern integrated circuit fabrication technologies, W is in the nanometer range.

The Emitter Current: The emitter current i_E is equal to the sum of the collector current i_C and the base current i ; that is $i_E = i_C + i_B$

$$i_E = \frac{\beta + 1}{\beta} i_C \qquad i_E = \frac{\beta + 1}{\beta} I_S e^{v_{BE}/V_T}$$

Alternatively, we can express i_E as $i_C = \alpha i_E$

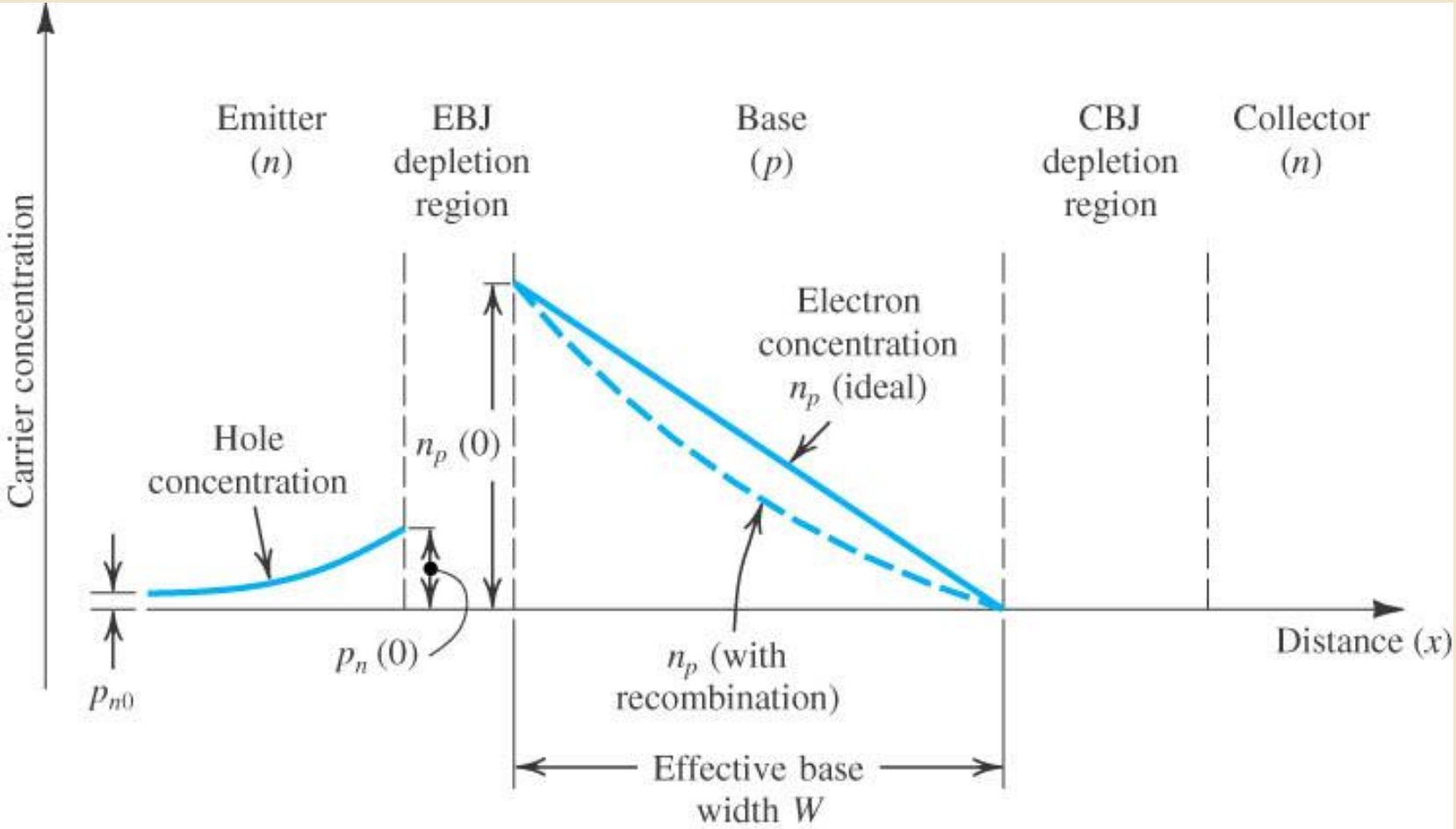
where the constant α is related to β by $\alpha = \frac{\beta}{\beta + 1}$

Thus the emitter current can also be written as $i_E = (I_S/\alpha) e^{v_{BE}/V_T}$

$$\beta = \frac{\alpha}{1 - \alpha}$$

α is called the common-base current gain

Minority-Carrier Distribution



Profiles of minority-carrier concentrations in the base and in the emitter of an npn transistor operating in the active mode: $v_{BE} > 0$ and $v_{CB} \geq 0$.

The concentration $n_p(0)$ will be proportional to e^{v_{BE}/V_T}

Similar to equation in p-n diode under forward biased diode, $p_n(x_n) = p_{n0}e^{V/V_T}$

Here, we have $n_p(0) = n_{p0}e^{v_{BE}/V_T}$ where n_{p0} is the thermal-equilibrium value of the minority-carrier (electron) concentration in the base region.

Because the base is very thin, the concentration of excess electrons decays almost linearly

The electron diffusion current I_n is directly proportional to the slope of the straight-line concentration profile,

$$I_n = A_E q D_n \frac{dn_p(x)}{dx} = A_E q D_n \left(-\frac{n_p(0)}{W}\right)$$

Where A_E is the cross-sectional area of the base–emitter unction, q is the magnitude of the electron charge, D_n Is the electron diffusivity in the base, and W is the effective width of the base. Finally, we have the collector current $i_C = I_n$

Collector Current $i_C = I_S e^{v_{BE}/V_T}$

The saturation current Is $I_S = A_E q D_n n_{p0} / W$

Substituting $n_{p0} = n_i^2 / N_A$, $I_S = \dfrac{A_E q D_n n_i^2}{N_A W}$

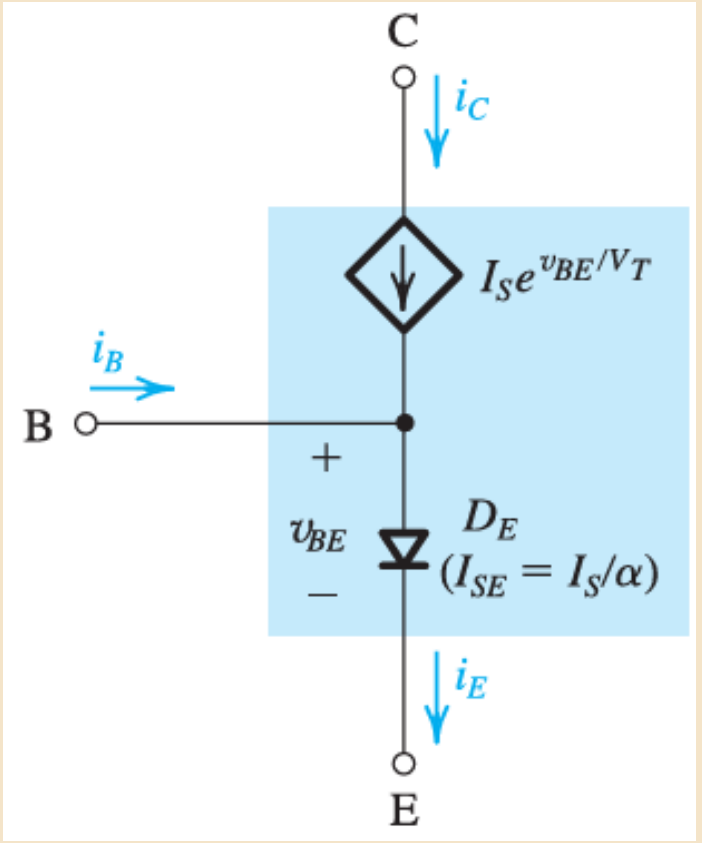
Here, the magnitude of i_C is independent of v+.
i.e, as long as the collector is positive with respect to the base, the electrons that reach the collector side of the base region will be swept into the collector and register as collector current.

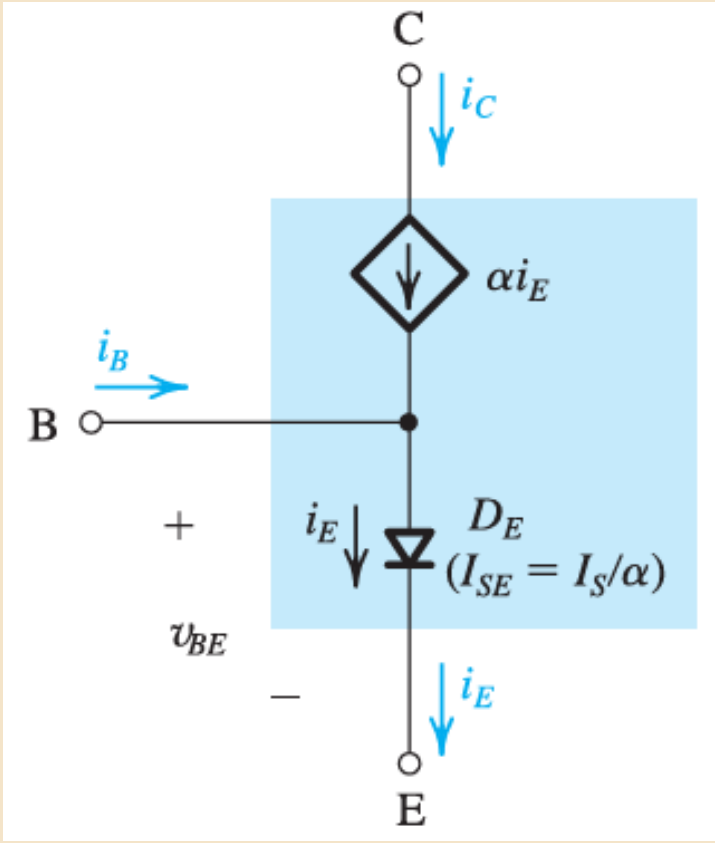
Equivalent circuit models

$$i_C = I_S e^{v_{BE}/V_T} \qquad i_E = (I_S/\alpha) e^{v_{BE}/V_T}$$

The current of the controlled source, which is equal to the collector current, is controlled by v_{BE}

$$i_C = \alpha i_E \qquad i_B = \frac{i_C}{\beta}$$





Here, diode D_E has a scale current I_{SE} equal to (I_S/α)

Fig. Large-signal equivalent-circuit models of the npn BJT operating in the forward active mode

It can be converted to the current-controlled current-source model shown in Fig. b by expressing the current of the controlled source as αi_E

Large-signal equivalent-circuit models of the *npn* BJT operating in the forward active mode

Equivalent circuit models

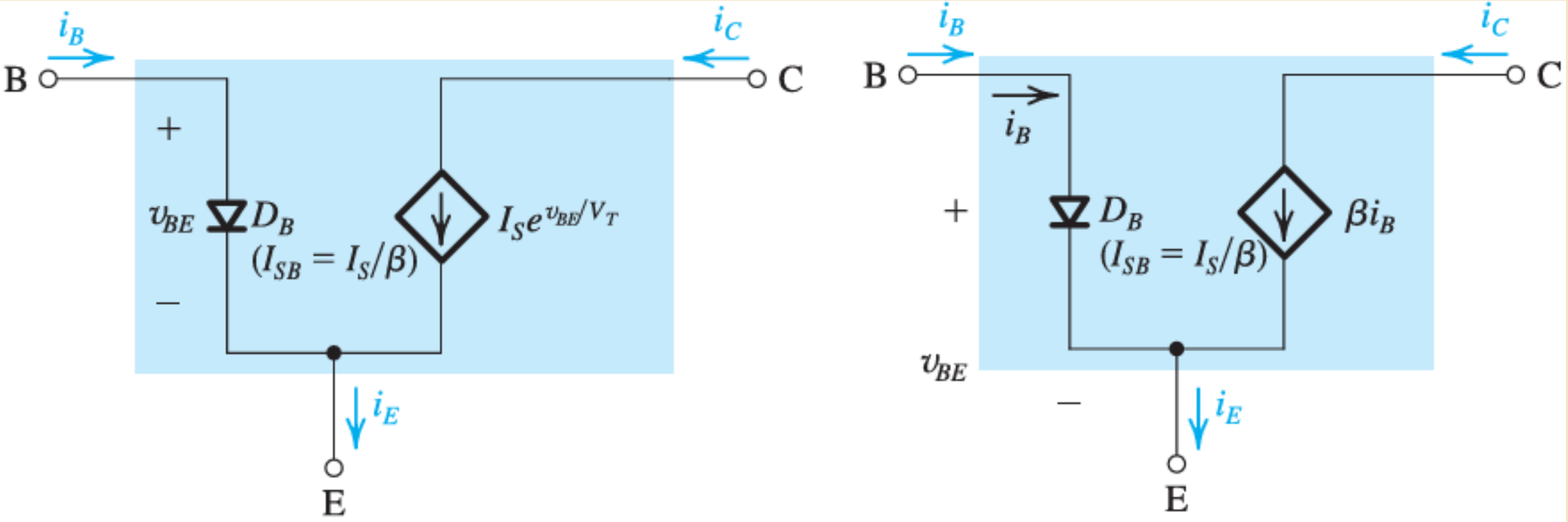


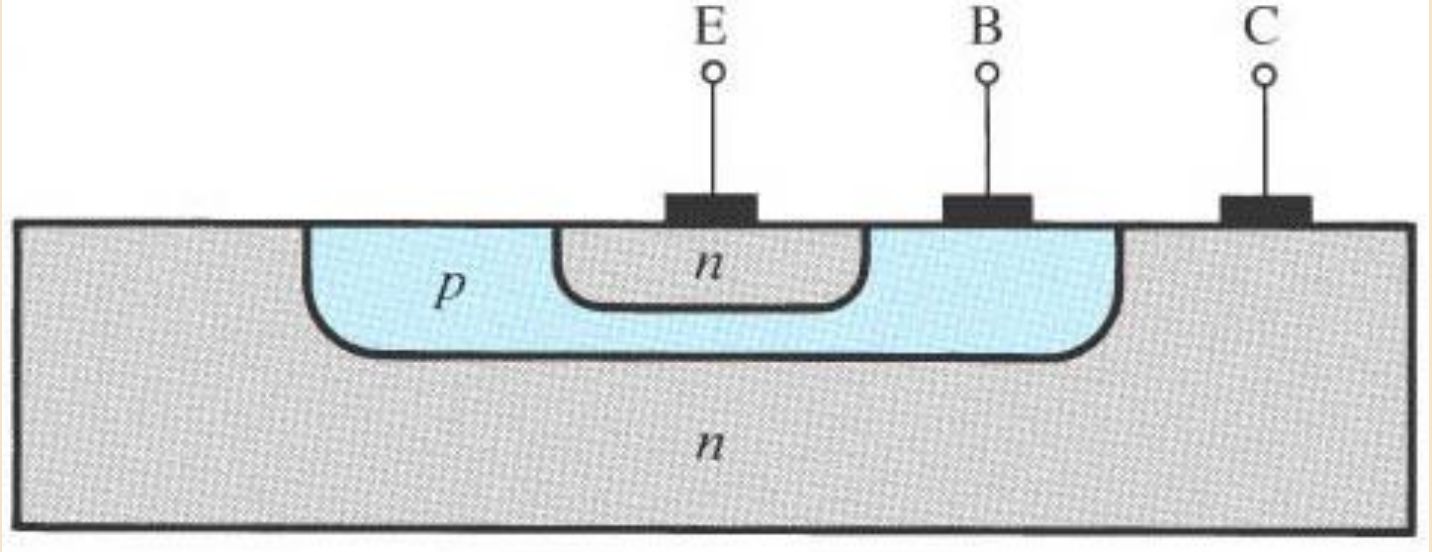
Fig. Large-signal equivalent-circuit models of the npn BJT operating in the forward active mode

$$i_B = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

The model of Fig.(c) is essentially a voltage-controlled current source. However, here diode D_B conducts the base current and thus its current scale factor is I_S/β

By simply expressing the collector current as βi_B we obtain the current-controlled current-source model shown in Fig.(d).

Structure of the Actual Transistors



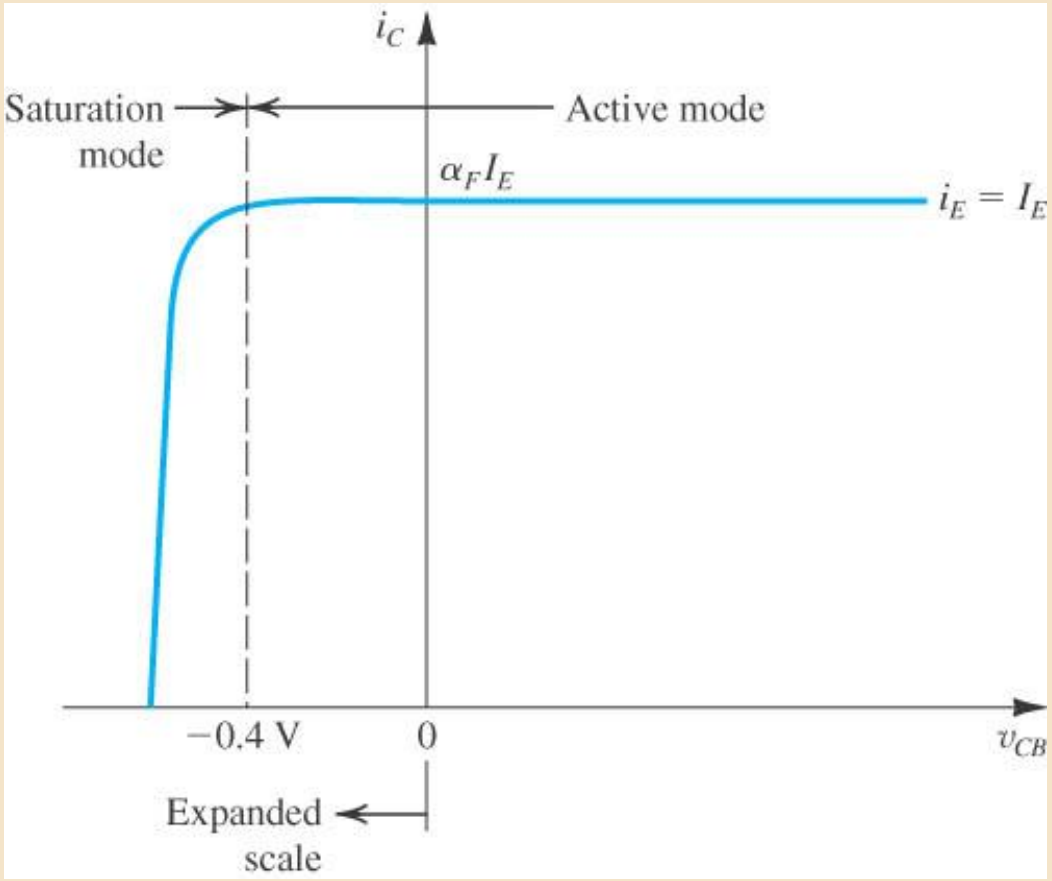
Cross-section of an *npn* BJT

The collector virtually surrounds the emitter region, thus making it difficult for the electrons injected into the thin base to escape being collected.

The device is not symmetrical, and thus the emitter and collector cannot be interchanged.

the CBJ has a much larger area than the EBJ. Thus the CB diode D_C has a saturation current I_{SC} that is much larger than the saturation current of the EB diode D_E . Typically, I_{SC} is 10 to 100 times larger than I_{SE} (recall that $I_{SE} = I_S / \alpha \simeq I_S$)

operation in Saturation Modes

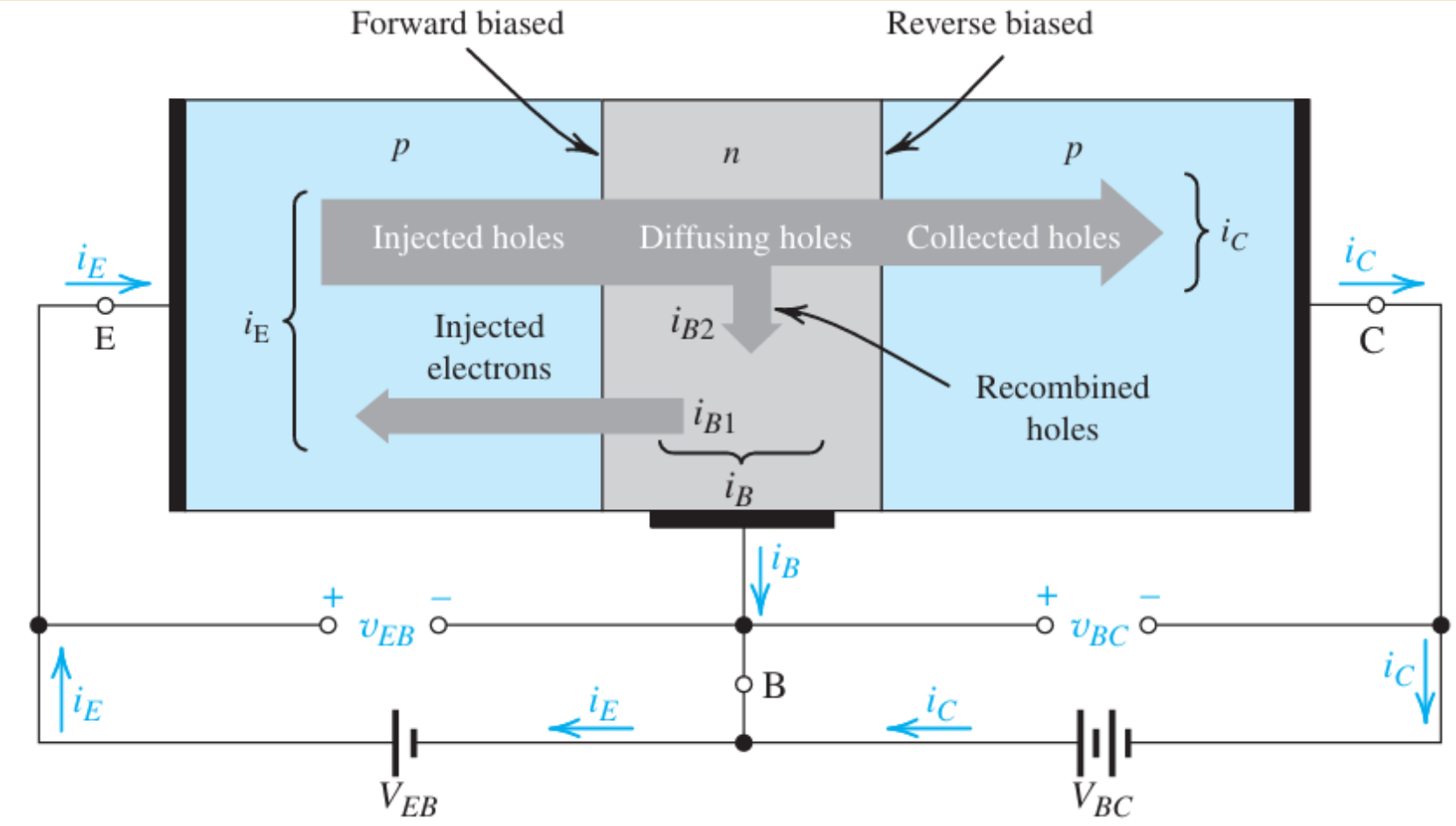


The i_C –v_{CB} characteristic of an *npn* transistor fed with a constant emitter current I_E. The transistor enters the saturation mode of operation for $v_{CB} < –0.4$ V, and the collector current diminishes.

When $v_{CB}>0$; the collector is at a higher potential than the base. Therefore, the collector-base junction is reverse biased and the BJT operates in active region.

When $v_{CB}<0.4V$; the collector is approximately 0.4 V below the base. Hence the CBJ junction is forward biased. The collector can no longer efficiently sweep out the carriers injected by the emitter. And i_C falls as v_{CB} becomes increasingly negative.

The PNP Transistor



Current flow in a pnp transistor biased to operate in the active mode

current in the pnp device is mainly conducted by holes injected from the emitter into the base as a result of the forward-bias voltage V_{EB}

Consider an *npn* transistor with $v_{BE} = 0.7$ V at $i_C = 1$ mA. Find v_{BE} at $i_C = 0.1$ mA and 10 mA.

Transistors of a certain type are specified to have β values in the range of 50 to 150. Find the range of their α values.

Find the output resistance of a BJT for which $V_A = 100$ V at $I_C = 0.1$, 1, and 10 mA

For a **silicon npn transistor**, assuming $V_{BE} = 0.7\,V$ at $I_C = 1\,mA$, we can use the BJT exponential relationship:

$$I_C = I_S e^{V_{BE}/V_T}$$

At constant temperature,

$$\frac{I_{C2}}{I_{C1}} = e^{(V_{BE2} - V_{BE1})/V_T}$$

Therefore,

$$V_{BE2} - V_{BE1} = V_T \ln\left(\frac{I_{C2}}{I_{C1}}\right)$$

At room temperature,

$$V_T \approx 25\,mV$$

1. For $I_C = 0.1\,mA$

Given:

$$I_{C1} = 1\,mA, \qquad V_{BE1} = 0.7\,V$$

$$V_{BE2} = 0.7 + 25mV \ln\left(\frac{0.1}{1}\right)$$

Since

$$\ln(0.1) = -2.303$$

$$V_{BE2} = 0.7 - (0.025)(2.303)$$

$$V_{BE2} = 0.7 - 0.0576$$

$$\boxed{V_{BE} \approx 0.642\,V}$$

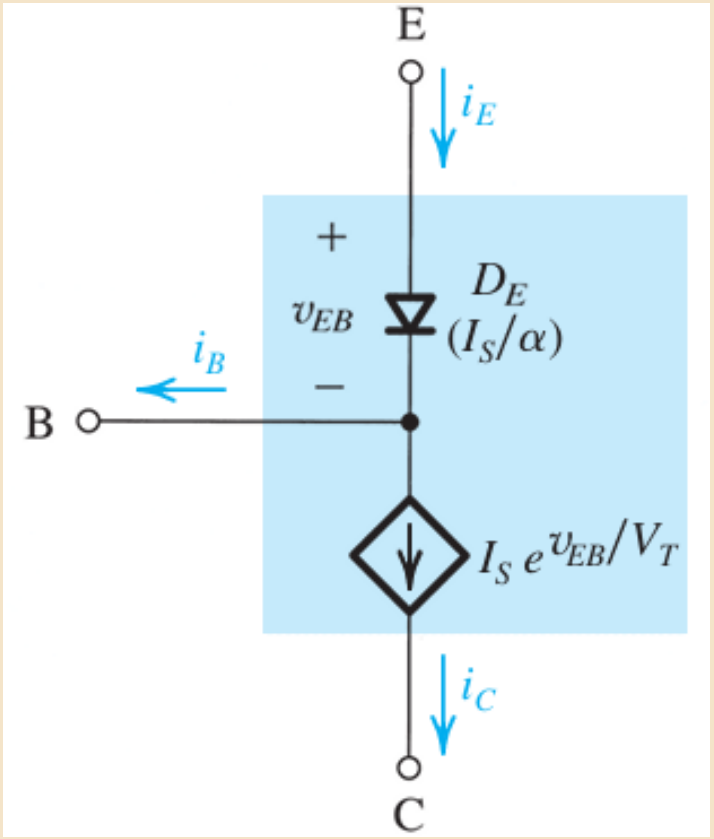
$$V_{BE2} = 0.7 + 25mV \ln\left(\frac{10}{1}\right)$$

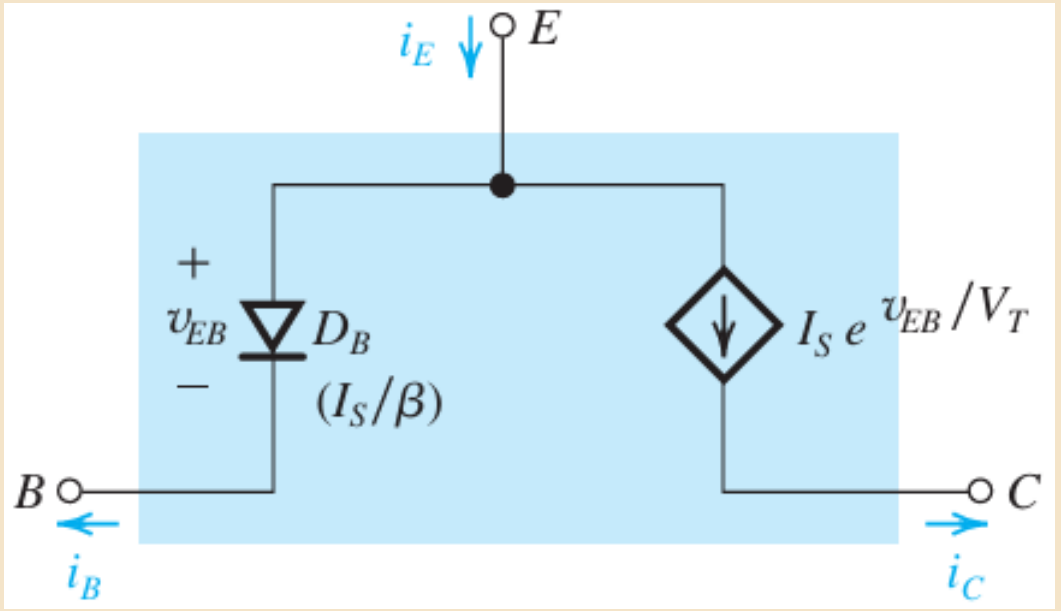
$$\ln(10) = 2.303$$

$$V_{BE2} = 0.7 + (0.025)(2.303)$$

$$V_{BE2} = 0.7 + 0.0576$$

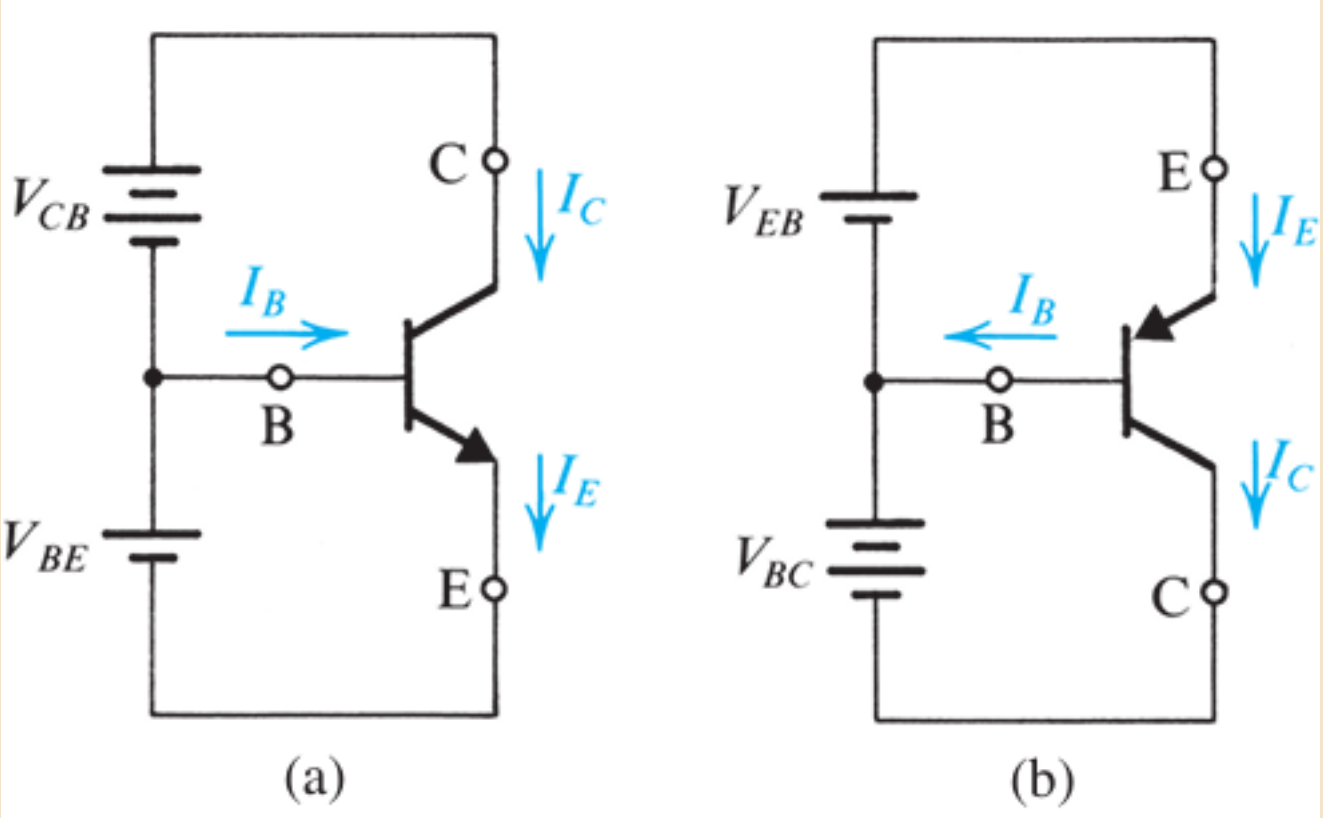
$$\boxed{V_{BE} \approx 0.758\,V}$$



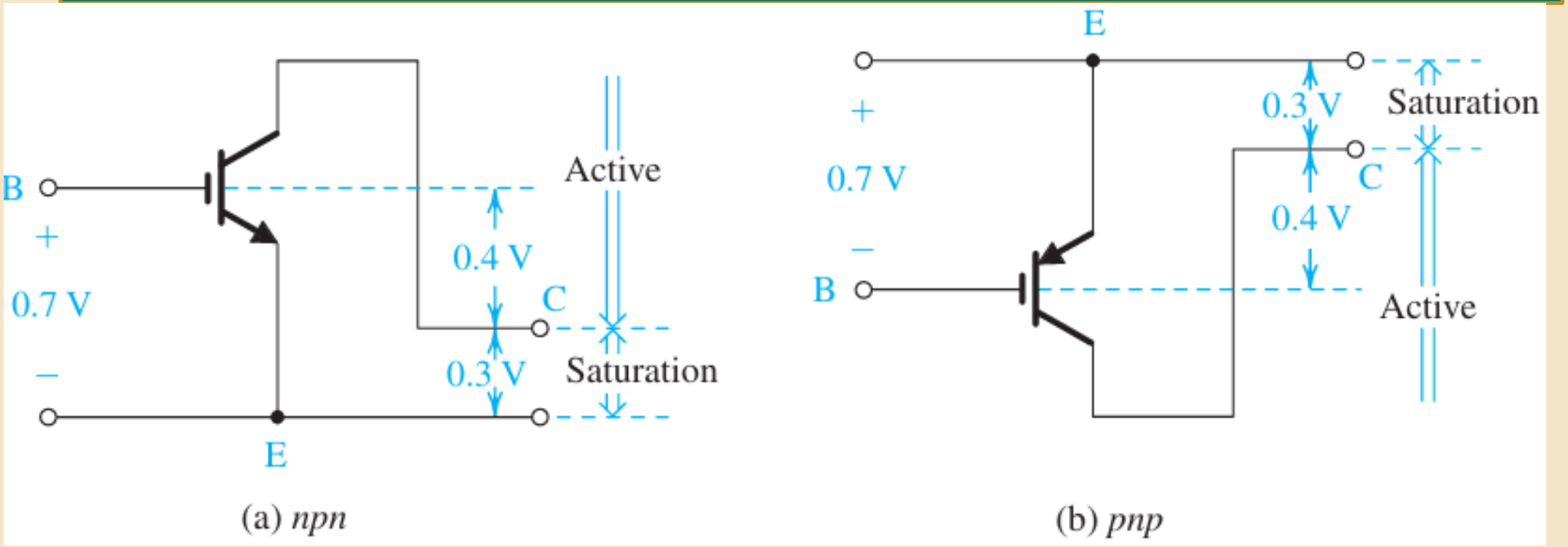


Two large-signal models for the pnp transistor operating in the active mode

Current–Voltage Characteristics



Voltage polarities and current flow in transistors operating in the active mode



Graphical representation of the conditions for operating the BJT in the active mode and in the saturation mode

The *npn* transistor whose EBJ is forward biased (usually, $V_{BE} \simeq 0.7$ V) will operate in the active mode as long as the collector voltage does not fall below that of the base by more than approximately 0.4V. Otherwise ,the transistor leaves the active mode and enters the saturation region of operation

The *pnp* transistor will operate in the active mode if the EBJ is forward biased (usually, $V_{EB} \simeq 0.7$ V) and the collector voltage is not allowed to rise above that of the base by more than 0.4 V or so. Otherwise, the CBJ becomes forward biased, and the *pnp* transistor enters the saturation region of operation

Summary of the BJT Current–Voltage Relationships in the Active Mode

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the *pnp* transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E \qquad i_B = (1-\alpha) i_E = \frac{i_E}{\beta+1}$$

$$i_C = \beta i_B \qquad i_E = (\beta+1) i_B$$

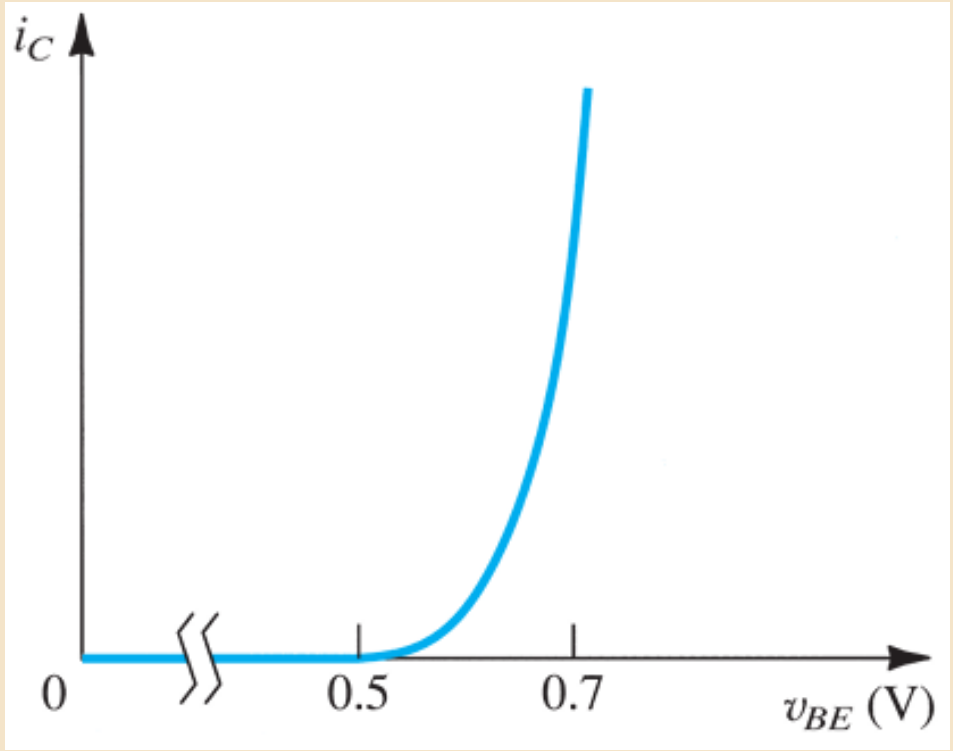
$$\beta = \frac{\alpha}{1-\alpha} \qquad \alpha = \frac{\beta}{\beta+1}$$

$$V_T = \text{thermal voltage} = \frac{kT}{q} \simeq 25 \text{ mV at room temperature}$$

The Collector–Base Reverse Current (I_{CBO})

The reverse current across the collector–base junction carried by thermally generated minority carriers is denoted $\mathbf{I_{CBO}}$. This is the reverse current flowing from collector to base with the emitter open-circuited (hence the subscript O). This current is usually in the nanoampere range, a value that is many times higher than its theoretically predicted value.

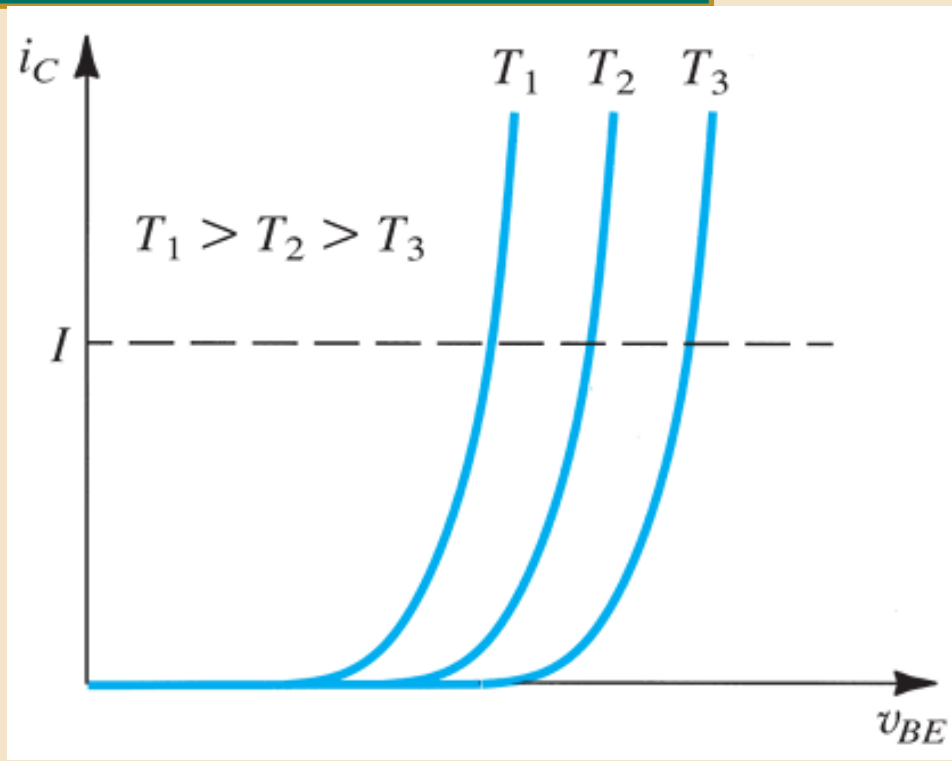
As with the diode reverse current, $\mathbf{I_{CBO}}$ contains a substantial leakage component, and its value is dependent on v_{CB}. $\mathbf{I_{CBO}}$ depends strongly on temperature, approximately doubling for every 10°C rise.



The iC–vBE characteristic for an npn transistor

$$i_C = I_S e^{v_{BE}/V_T}$$

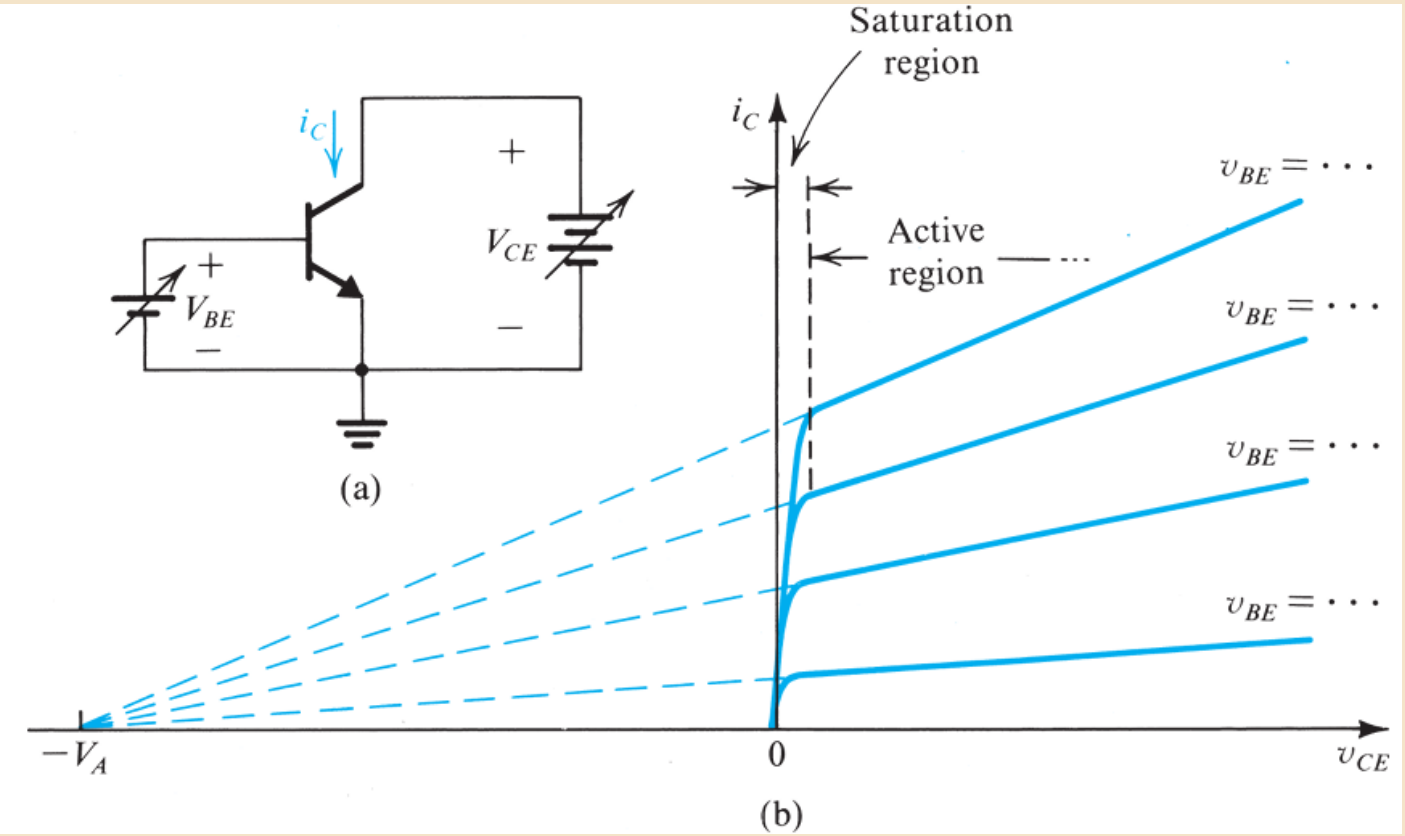
For a pnp transistor, the i_C–v_{EB} characteristic will look identical to this figure with v_{BE} replaced with v_{EB}



Effect of temperature on the iC–vBE characteristic. At a constant emitter current (broken line), vBE changes by −2mV/°C

As in silicon diodes, the voltage across the emitter–base junction decreases by about 2 mV for each rise of 1°C in temperature, provided the junction is operating at a constant current

Dependence of i_C on the Collector Voltage—The Early Effect



(a) Conceptual circuit for measuring the i_C–v_{CE} characteristics of the BJT
(b) The i_C–v_{CE} characteristics of a practical BJT

$$V_{CE} \uparrow \Rightarrow V_{CB} \uparrow \Rightarrow W_B \downarrow \Rightarrow I_C \uparrow$$

Early voltage V_A is the magnitude of the negative V_{CE}-axis intercept obtained by extrapolating the BJT output characteristics, and it represents the effect of base-width modulation on collector current.

The effect was identified and explained by **James M. Early**, an American semiconductor engineer who worked at Bell Telephone Laboratories. He studied the phenomenon of **base-width modulation** in bipolar transistors

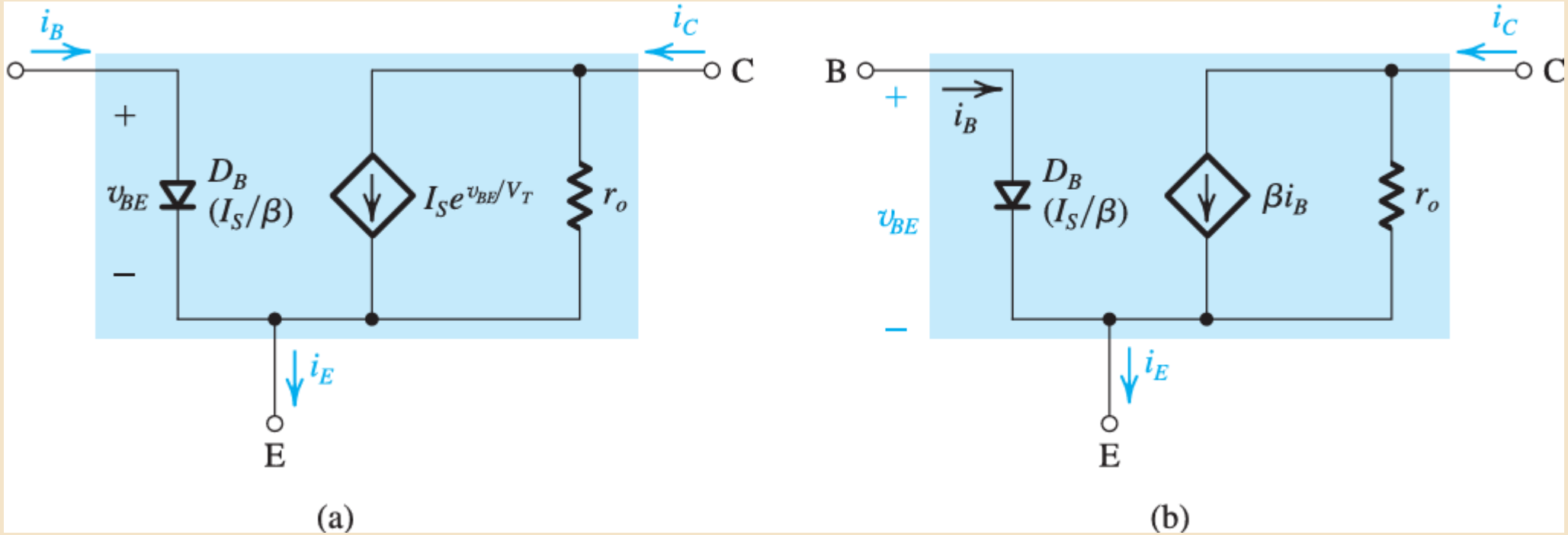
$$i_C = I_S e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right)$$

output resistance looking into the collector is not infinite.

typical values of V_A lies in the range of 10 V to 100 V

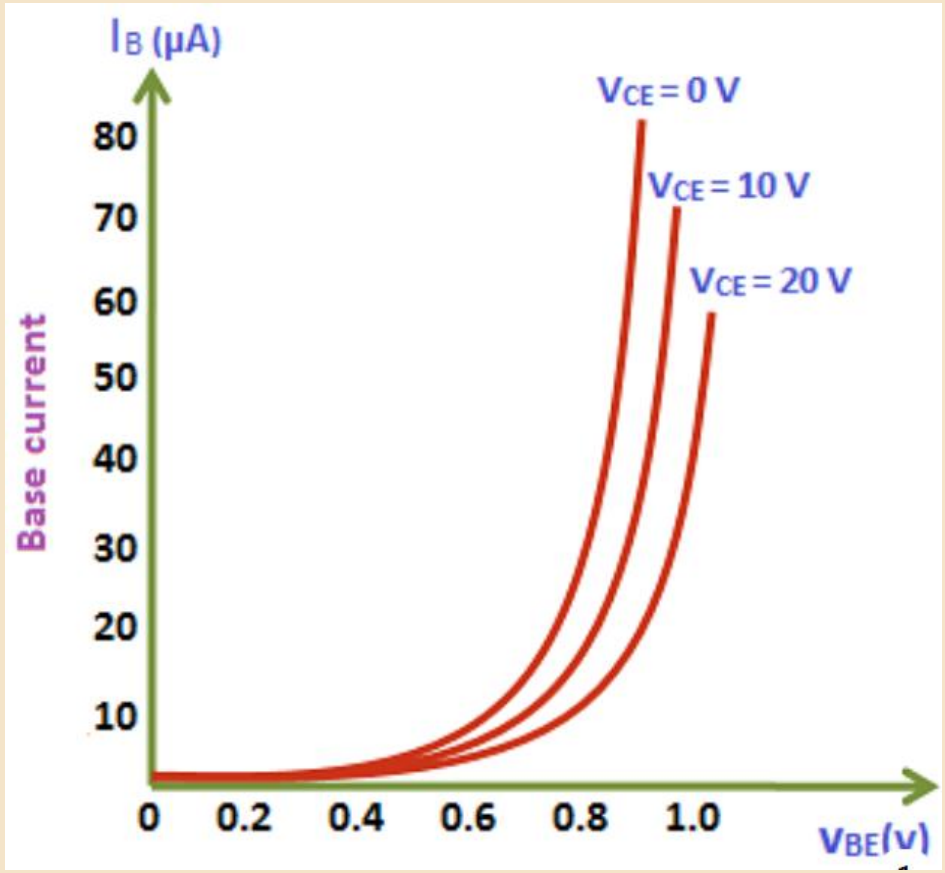
$$r_o = \frac{V_A}{I'_C}$$

$$I'_C = I_S e^{V_{BE}/V_T}$$

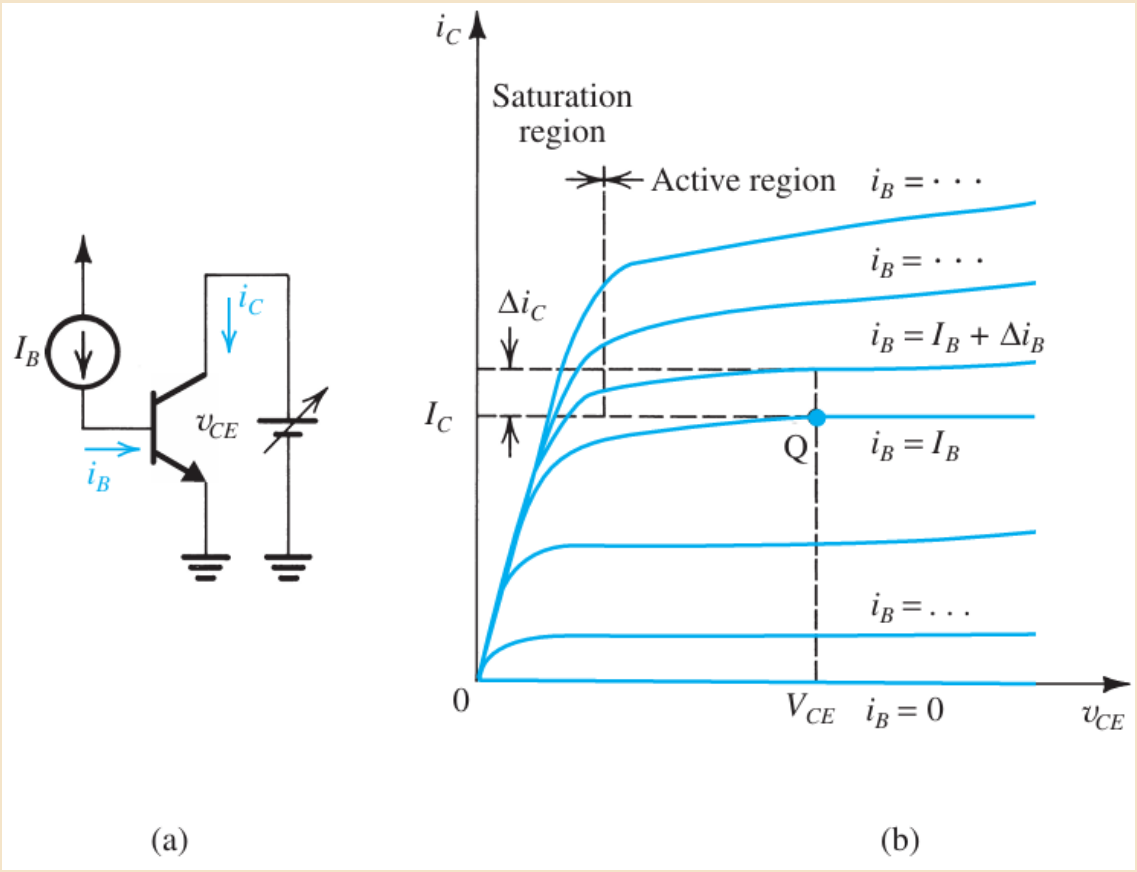


Large-signal, equivalent-circuit models of an npn BJT operating in the active mode in the common-emitter configuration with the output resistance r_o included.

Common-Emitter Input and Output Characteristics

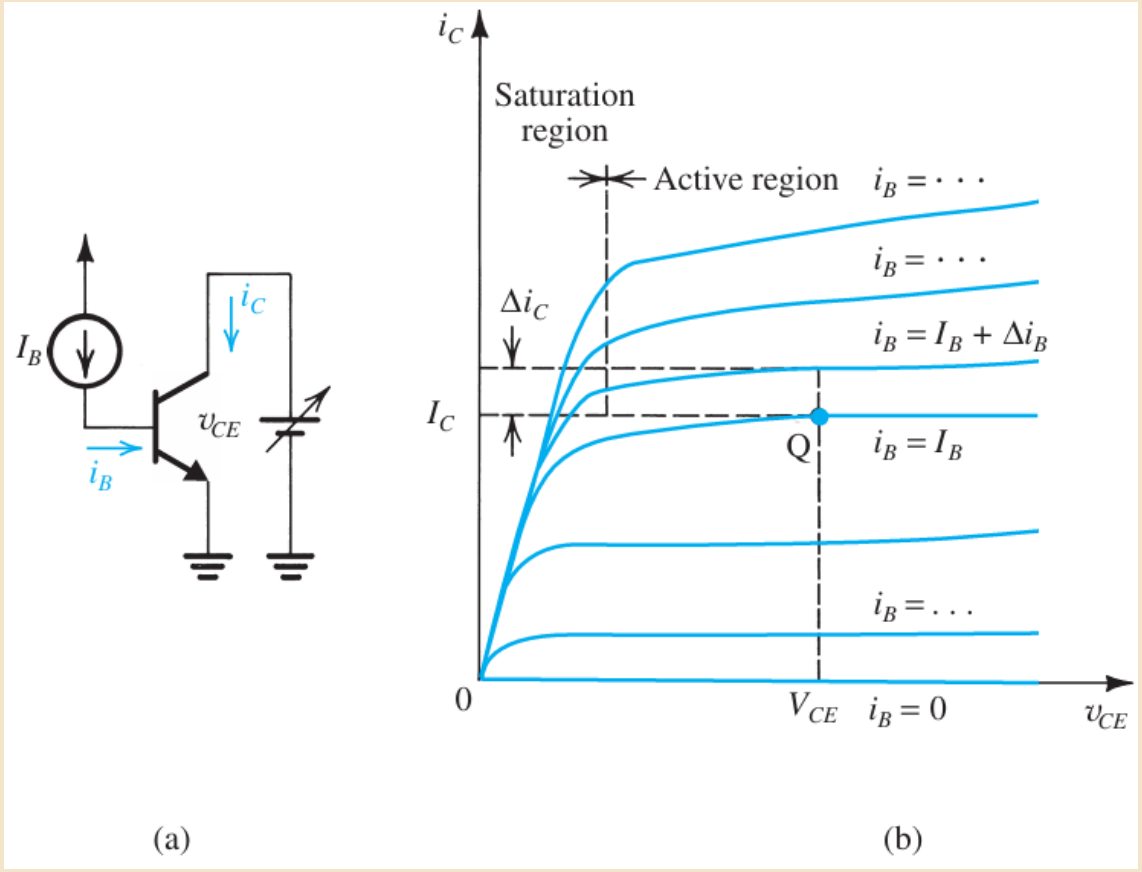


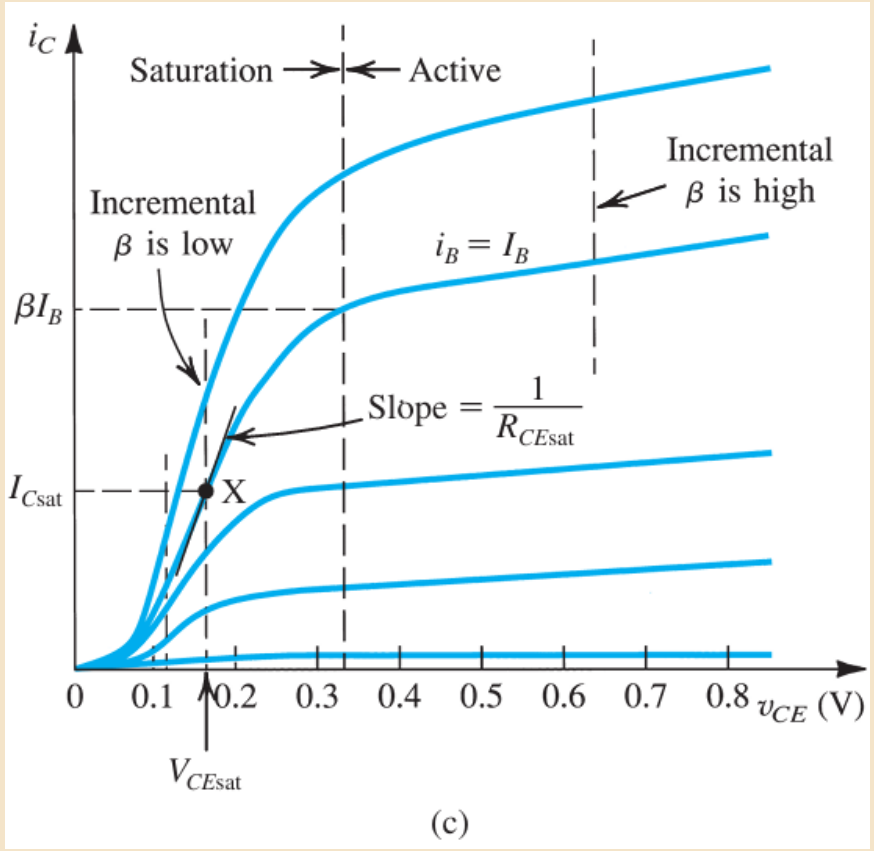
Common-Emitter input characteristics



Common-Emitter output characteristics

An Alternative Form of the Common-Emitter Characteristics





Common-emitter characteristics. (a) Basic CE circuit; note that in (b) the horizontal scale is expanded around the origin to show the saturation region in some detail. A much greater expansion of the saturation region is shown in (c)

Operating point in active region is Q

The transistor β is the ratio of I_C / I_B or $\Delta i_C / \Delta i_B$

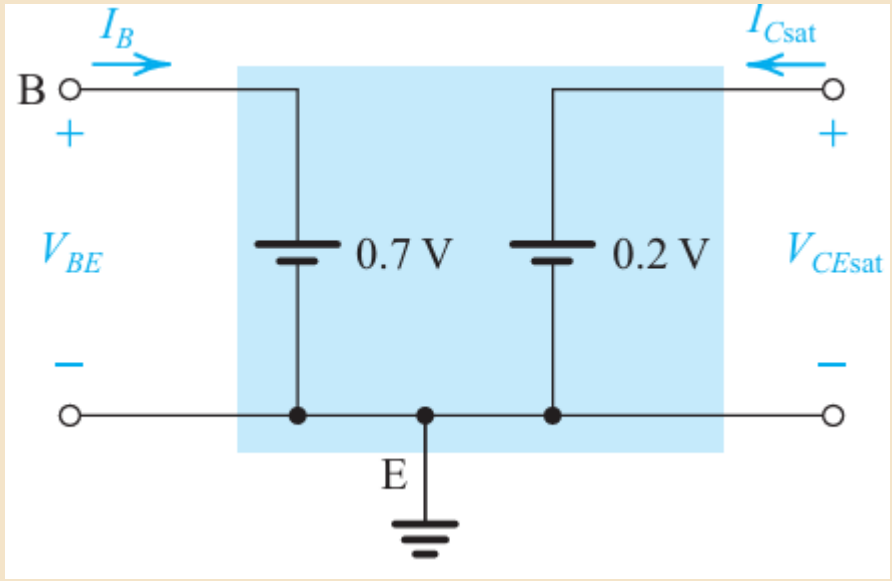
Operating point in Saturation region is X

The incremental β is lower there than in the active region

$$\beta_{\text{forced}} = \left.\frac{i_C}{i_B}\right|_{\text{saturation}} \leq \beta$$

$$R_{CEsat} \equiv \left.\frac{\partial v_{CE}}{\partial i_C}\right|_{\substack{i_B = I_B \\ i_C = I_{Csat}}}$$

Typically, RCE_{sat} ranges from a few ohms to a few tens of ohms



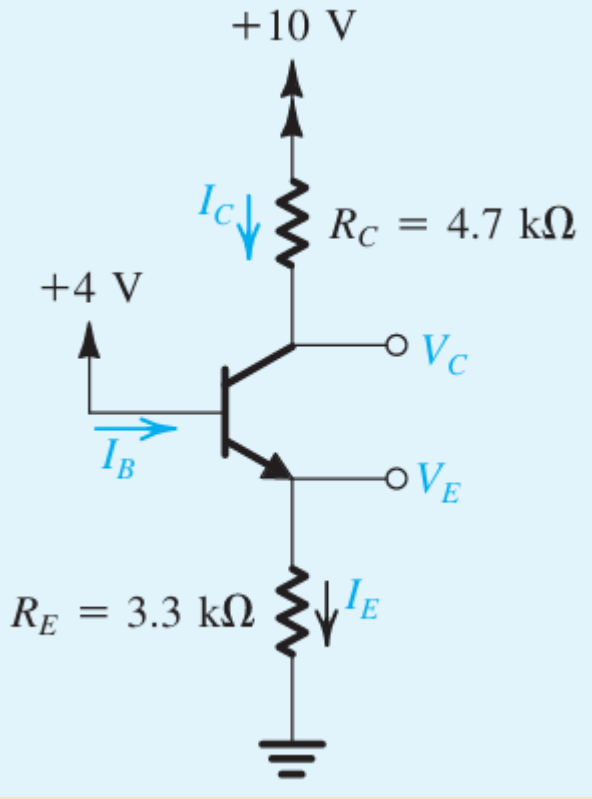
A simplified equivalent-circuit model of the saturated transistor

$|V_{CE}|$ of a saturated transistor is 0.2 V

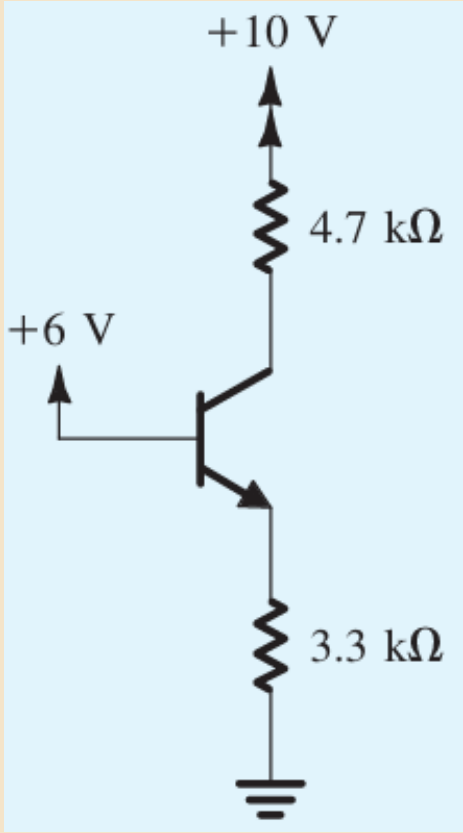
BJT Circuits at DC

Analysis of BJT circuits to which only dc voltages are applied

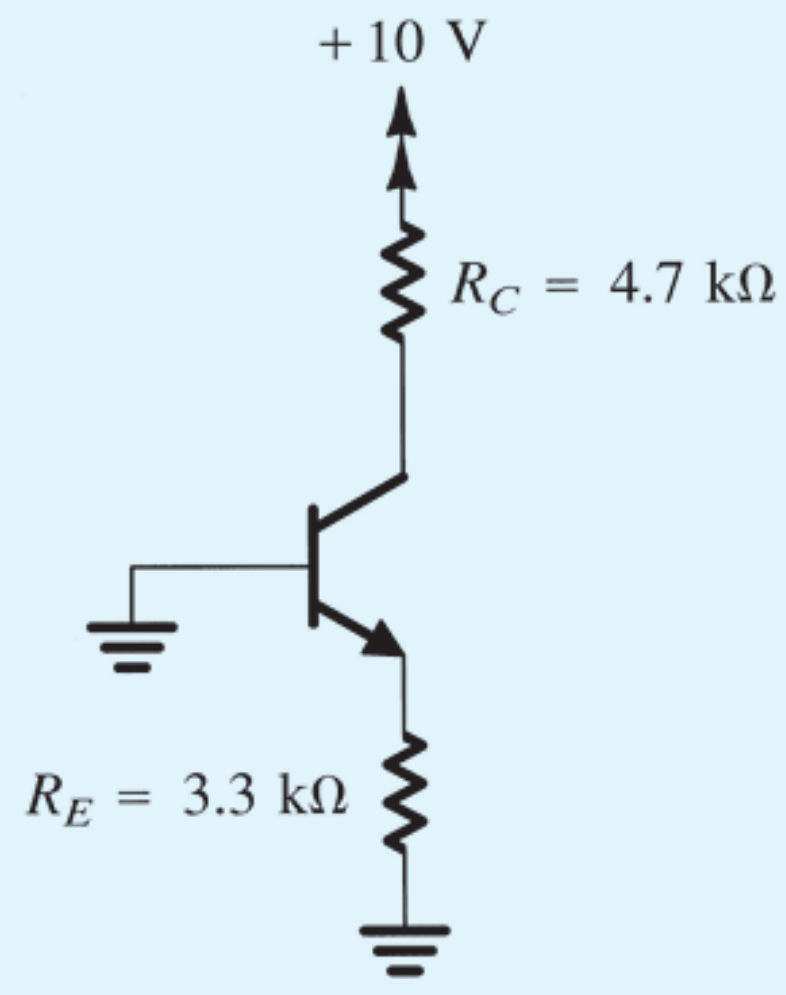
Consider the circuit shown in Fig. Analyze this circuit to determine all node voltages and branch currents. Assume that β is specified to be 100.



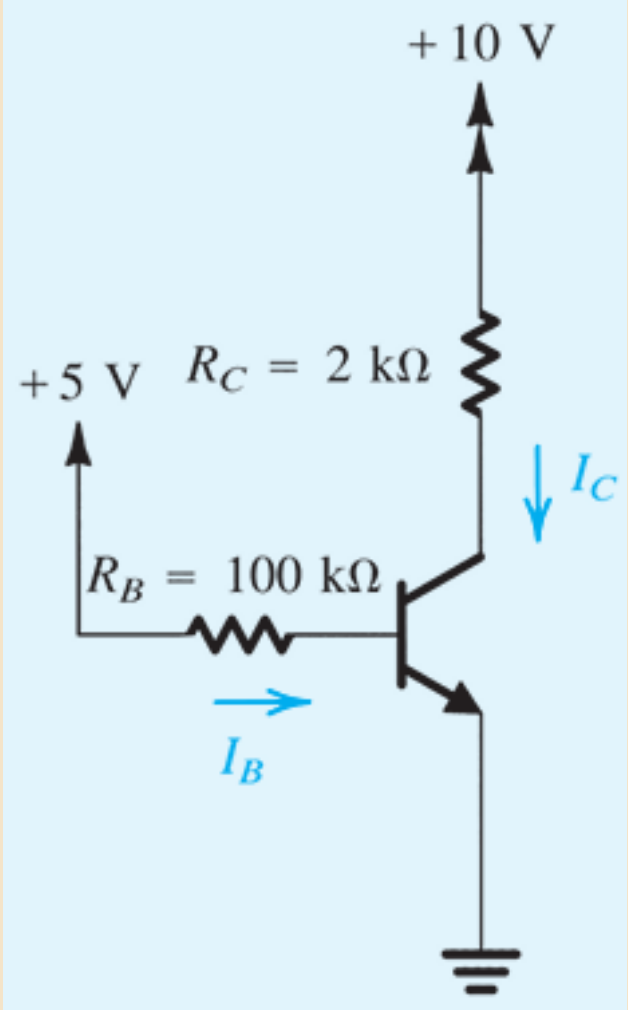
Consider the circuit shown in Fig. Analyze this circuit to determine all node voltages and branch currents. Assume that β is specified to be 50.



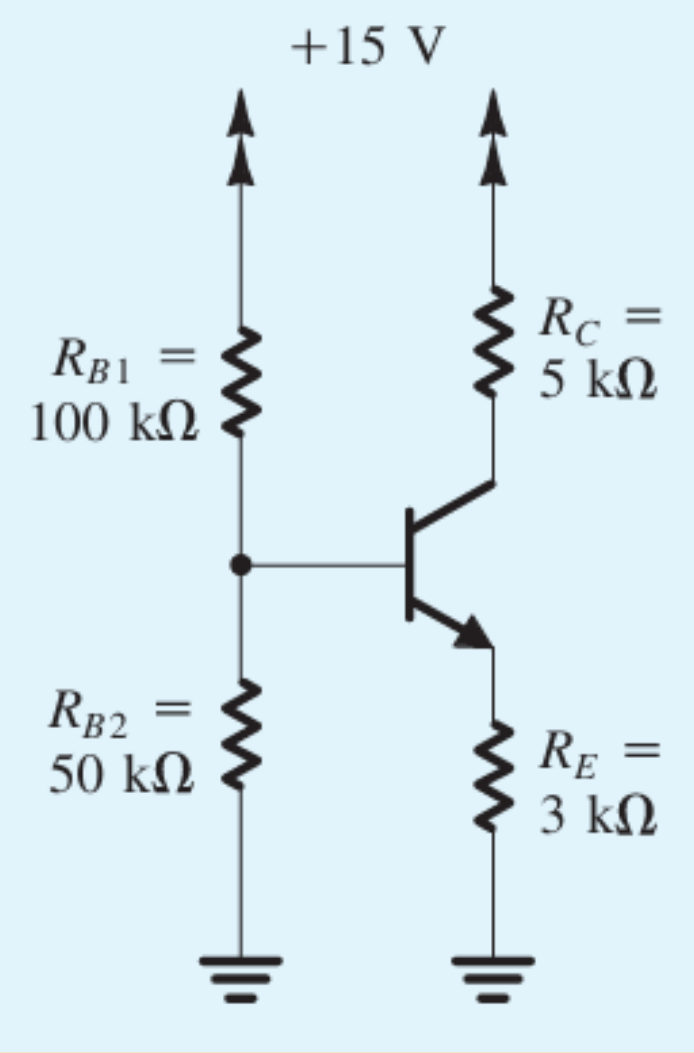
Analyze the circuit. Determine the voltages at all nodes and the currents through all branches



Analyze the circuit. Determine the voltages at all nodes and the currents through all branches. Assume β = 100

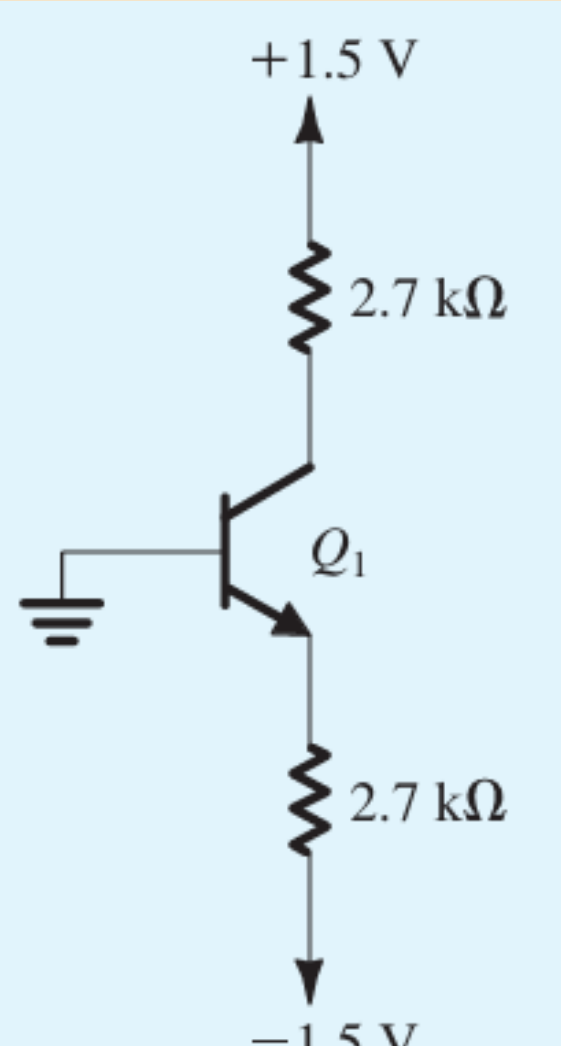


Analyze the circuit. Determine the voltages at all nodes and the currents through all branches. Assume β = 100

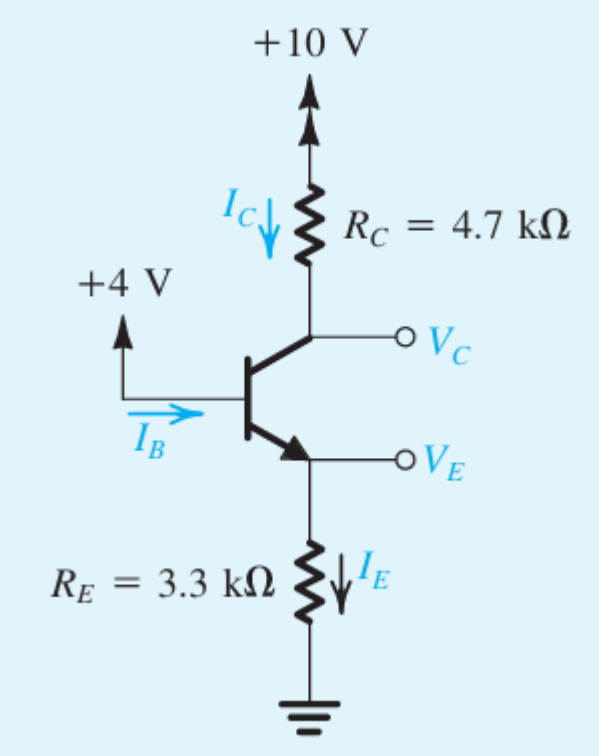


If the transistor in the circuit is replaced with another having half the value of β (i.e., β=50), find the new value of I_C , and express the change in I_C as a percentage.

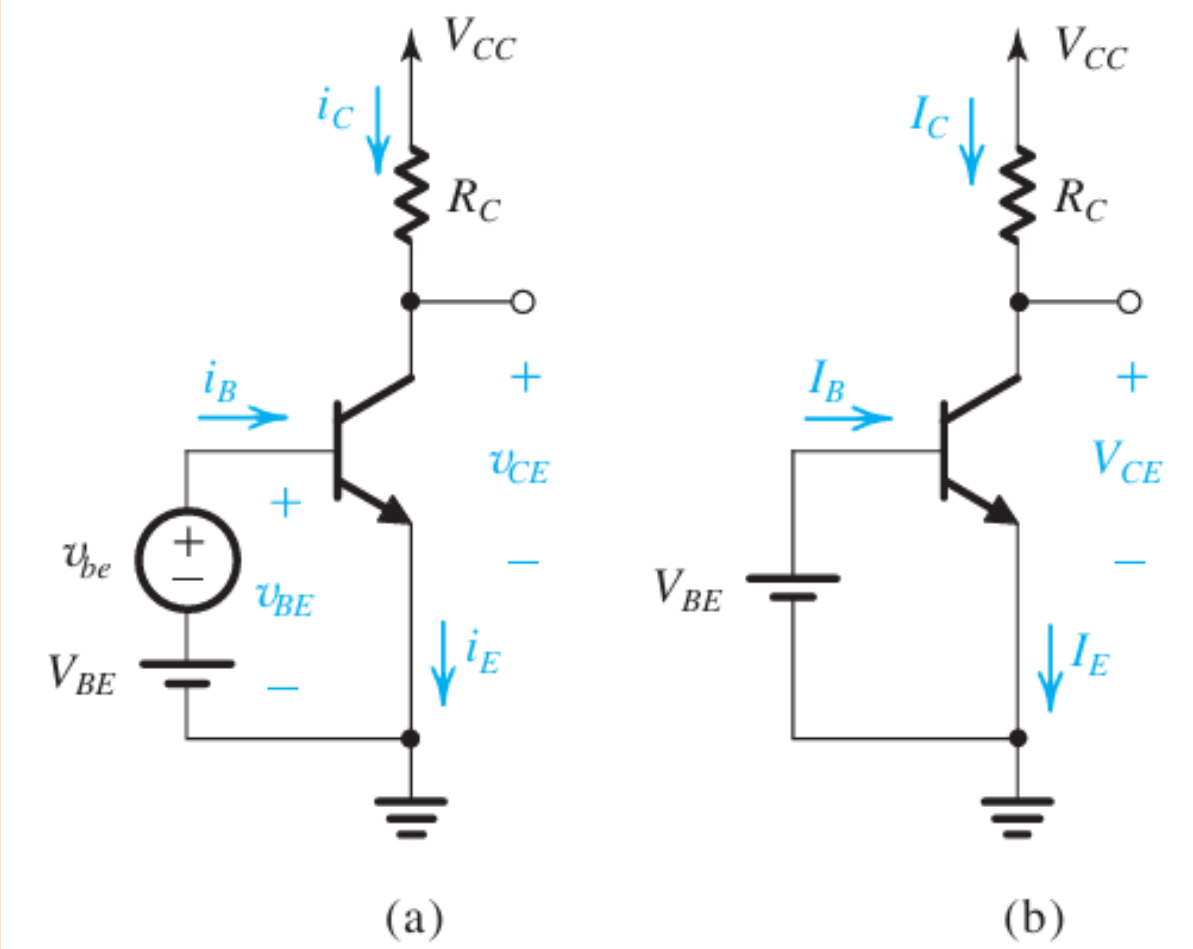
find the emitter, base, and collector voltages and currents. Use β =50, but assume V_{BE} = 0.8 V independent of current level.



Redesign the circuit i.e. find new values for R_E and R_C to establish a collector current of 0.5 mA and a reverse-bias voltage on the collector–base junction of 2 V. Assume $\alpha \simeq 1$



Small-Signal Operation and Models



$$I_C = I_S e^{V_{BE}/V_T}$$
$$I_E = I_C/\alpha$$
$$I_B = I_C/\beta$$
$$V_{CE} = V_{CC} - I_C R_C$$

For active-mode operation, V_{CE} should be greater than(V_{BE} −0.4) by an amount that allows for the required negative signal swing at the collector.

(a) Conceptual circuit to illustrate the operation of the transistor as an amplifier.
(b) The circuit of (a) with the signal source v_{be} eliminated for dc (bias) analysis.

The Collector Current and the Transconductance

If a signal v_{be} is applied as shown in Fig. (a), the total instantaneous base–emitter voltage v_{BE} becomes

$$v_{BE} = V_{BE} + v_{be}$$

Correspondingly, the collector current becomes

$$i_C = I_S e^{v_{BE}/V_T} = I_S e^{(V_{BE}+v_{be})/V_T}$$
$$= I_S e^{V_{BE}/V_T} e^{v_{be}/V_T}$$

Therefore $i_C = I_C e^{v_{be}/V_T}$

Now, if $v_{be} \ll V_T$, we may approximate the above equation as $i_C \simeq I_C\left(1 + \frac{v_{be}}{V_T}\right)$ is the **small-signal approximation** for the BJT

The small-signal approximation for the BJT is valid only for vbe less than 5 mV or 10mV, at most.
Under this approximation, the total collector current is given by above equation can be rewritten $i_C = I_C + \frac{I_C}{V_T} v_{be}$

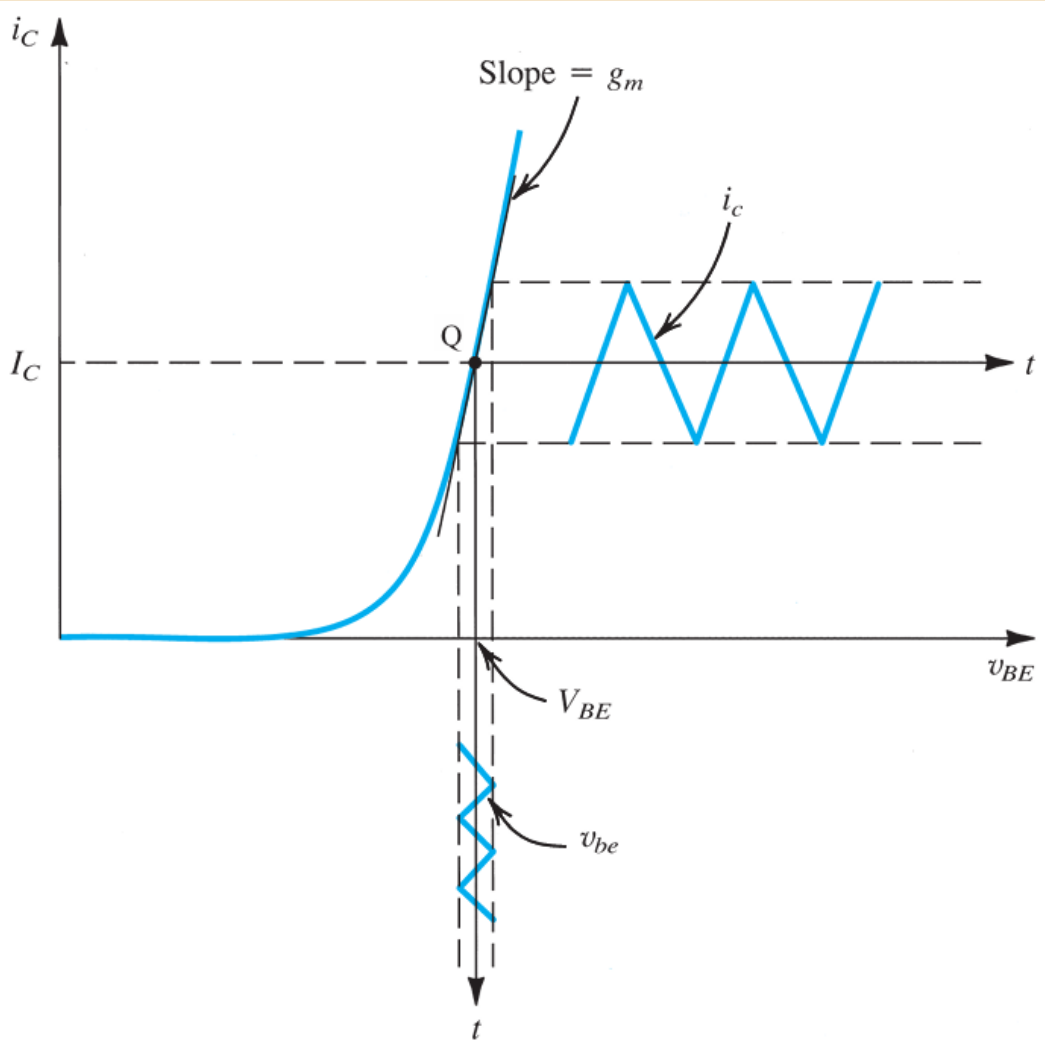
Thus the collector current is composed of the dc bias value I_C and a signal component i_c, $i_c = \frac{I_C}{V_T} v_{be}$

This equation relates the signal current in the collector to the corresponding base–emitter signal voltage. It can be rewritten as $i_c = g_m v_{be}$

where g_m is the transconductance, and is given by $g_m = \frac{I_C}{V_T}$

the transconductance of the BJT is directly proportional to the collector bias current I_C

At I_C =1 mA, $g_m \simeq$ 40mA/V



A graphical interpretation for gm is given in Fig, where it is shown that g_m is equal to the slope of the tangent to the i_C–v_{BE} characteristic curve at $i_C = I_C$ (i.e., at the bias point Q).

$$g_m = \left.\frac{\partial i_C}{\partial v_{BE}}\right|_{i_C = I_C}$$

The small-signal approximation implies keeping the signal amplitude sufficiently small that operation is restricted to an almost-linear segment of the i_C–v_{BE} exponential curve. Increasing the signal amplitude will result in the collector current having components nonlinearly related to v_{be}

Linear operation of the transistor under the small-signal condition: A small-signal v_{be} with a triangular waveform is superimposed on the dc voltage V_{BE} . It gives rise to a collector-signal current i_c , also of triangular waveform, superimposed on the dc current I_C. Here, $i_c = g_m v_{be}$, where g_m is the slope of the the i_C–v_{BE} curve at the bias point Q.

The Base Current and the Input Resistance at the Base

To determine the resistance seen by v_{be} , we first evaluate the total base current i_B

$$i_B = \frac{i_C}{\beta} = \frac{I_C}{\beta} + \frac{1}{\beta}\frac{I_C}{V_T}v_{be} \quad \Rightarrow \quad i_B = I_B + i_b$$

where I_B is equal to I_C/β and the signal component i_b is given by $i_b = \frac{1}{\beta}\frac{I_C}{V_T}v_{be}$

Substituting for I_C/V_T by g_m gives $i_b = \frac{g_m}{\beta}v_{be}$

The small-signal input resistance between base and emitter, looking into the base, is denoted by r_π and is defined as

$$r_\pi \equiv \frac{v_{be}}{i_b} \quad \Rightarrow \quad r_\pi = \frac{\beta}{g_m} \quad \Rightarrow \quad r_\pi = \frac{V_T}{I_B}$$

The Emitter Current and the Input Resistance at the Emitter

$$i_C = I_C + \frac{I_C}{V_T} v_{be}$$

The total emitter current, $i_E = \frac{i_C}{\alpha} = \frac{I_C}{\alpha} + \frac{i_c}{\alpha}$

$$i_E = I_E + i_e \Rightarrow i_e = \frac{i_c}{\alpha} = \frac{I_C}{\alpha V_T} v_{be} = \frac{I_E}{V_T} v_{be}$$

If we denote the small-signal resistance between base and emitter looking into the emitter by r_e , it can be defined as

$$r_e \equiv \frac{v_{be}}{i_e}$$

From the above equation, the **emitter resistance**, r_e, is given by $r_e = \frac{V_T}{I_E}$

Considering the equation for transconductance $g_m = \frac{I_C}{V_T}$, the equation for r_e can be given as $r_e = \frac{\alpha}{g_m} \simeq \frac{1}{g_m}$

Considering the equations for r_π and r_e, i.e. $r_\pi \equiv \frac{v_{be}}{i_b}$ and $r_e \equiv \frac{v_{be}}{i_e}$

$$v_{be} = i_b r_\pi = i_e r_e$$

Therefore, $r_\pi = (i_e/i_b) r_e \Rightarrow r_\pi = (\beta + 1) r_e$

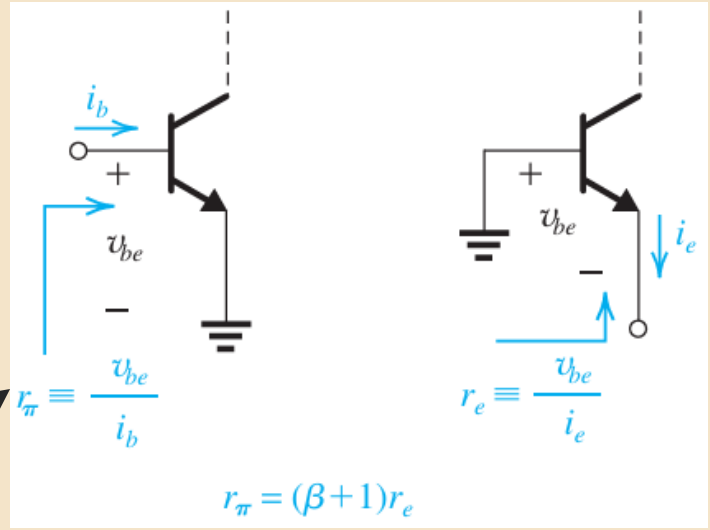


Fig. Illustrating the definition of r_π and r_e

The Voltage Gain

The total collector voltage v_{CE} is

$$\begin{aligned} v_{CE} &= V_{CC} - i_C R_C \\ &= V_{CC} - (I_C + i_c) R_C \\ &= (V_{CC} - I_C R_C) - i_c R_C \\ &= V_{CE} - i_c R_C \end{aligned}$$

Thus, superimposed on the collector bias voltage V_{CE} we have signal voltage v_{ce} given by

$$\begin{aligned} v_{ce} &= -i_c R_C = -g_m v_{be} R_C \\ &= (-g_m R_C) v_{be} \end{aligned}$$

From which we find the voltage gain A_v of this amplifier as

$$A_v \equiv \frac{v_{ce}}{v_{be}} = -g_m R_C$$

Here again we note that because gm is directly proportional to the collector bias current, the gain will be as stable as the collector bias current is made.

$$A_v = -\frac{I_C R_C}{V_T}$$

We make the following observations on this expression for the voltage gain:

(i) The gain is negative, which signifies that the amplifier is inverting; that is, there is a 180° phase shift between the input and the output.

(ii) The gain is proportional to the collector bias current I_C and to the load resistance R_C.

Separating the Signal and the DC Quantities

Representation of the signal operation of the BJT can be obtained by eliminating the dc sources.

To perform ac analysis, open all DC currents and short all DC sources. It is not an actual amplifier circuit, since the dc bias circuit is not shown. it can be considered an equivalent small-signal circuit model.

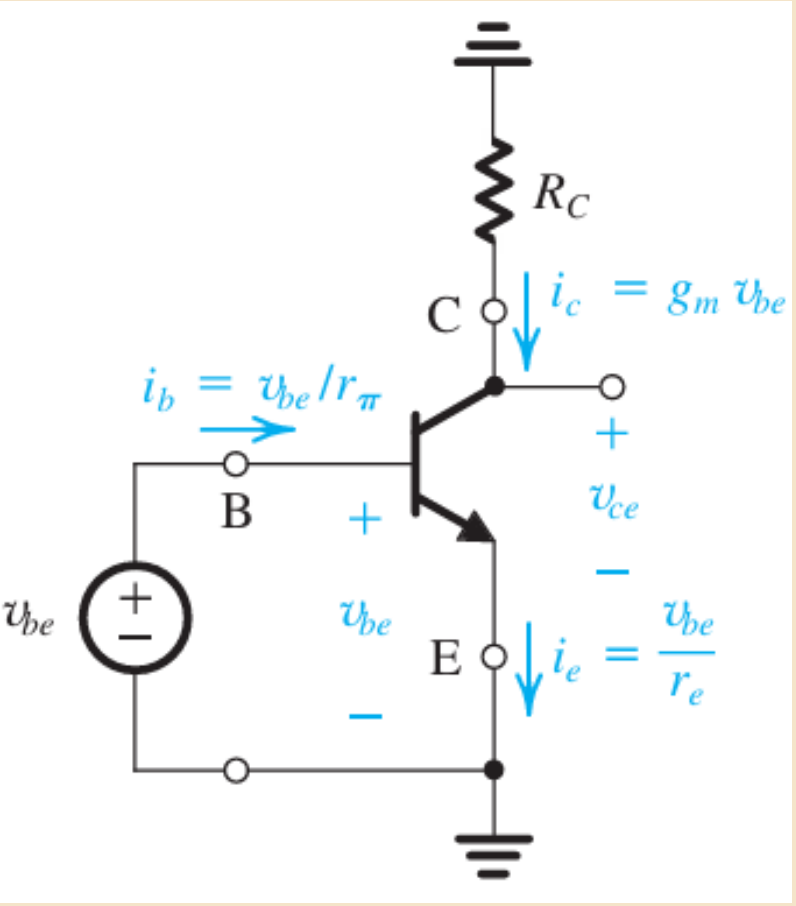
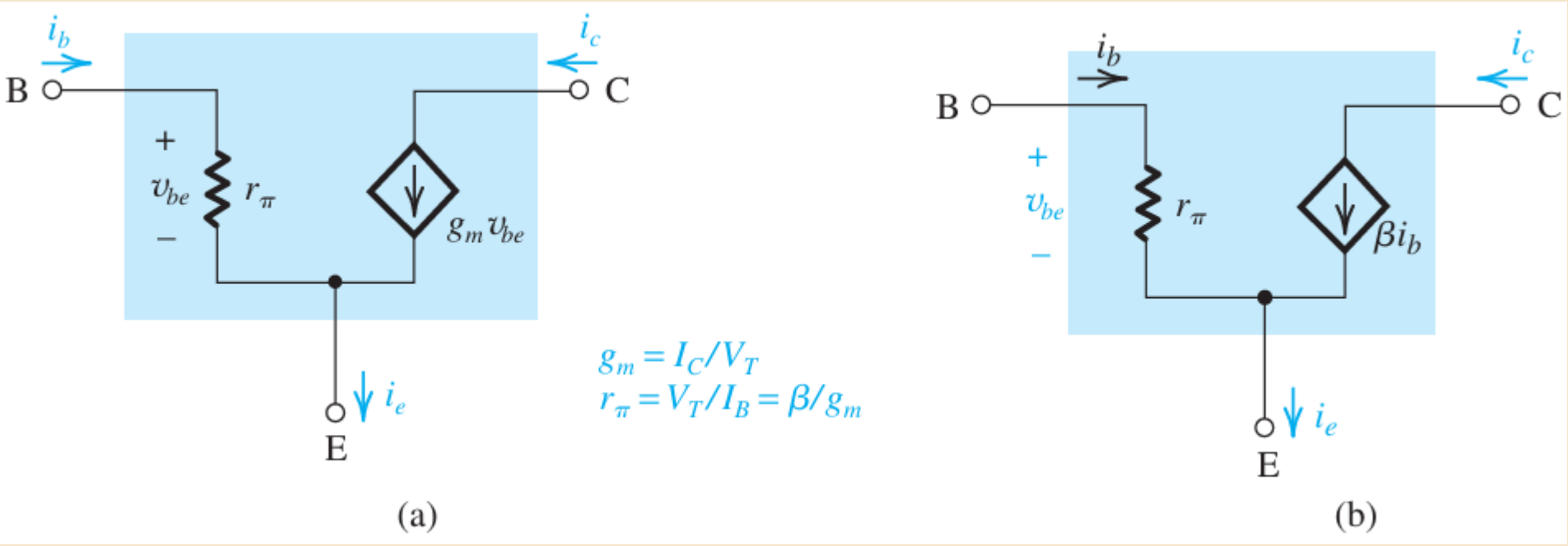


Figure shows the expressions for the current increments (i_c, i_b, and i_e) obtained when a small signal v_{be} is applied.

The Hybrid-π Model



$$i_c = g_m v_{be}$$

$$r_\pi \equiv \frac{v_{be}}{i_b}$$

$$r_\pi = \frac{\beta}{g_m}$$

$$r_e \equiv \frac{v_{be}}{i_e}$$

$$r_\pi = (\beta + 1) r_e$$

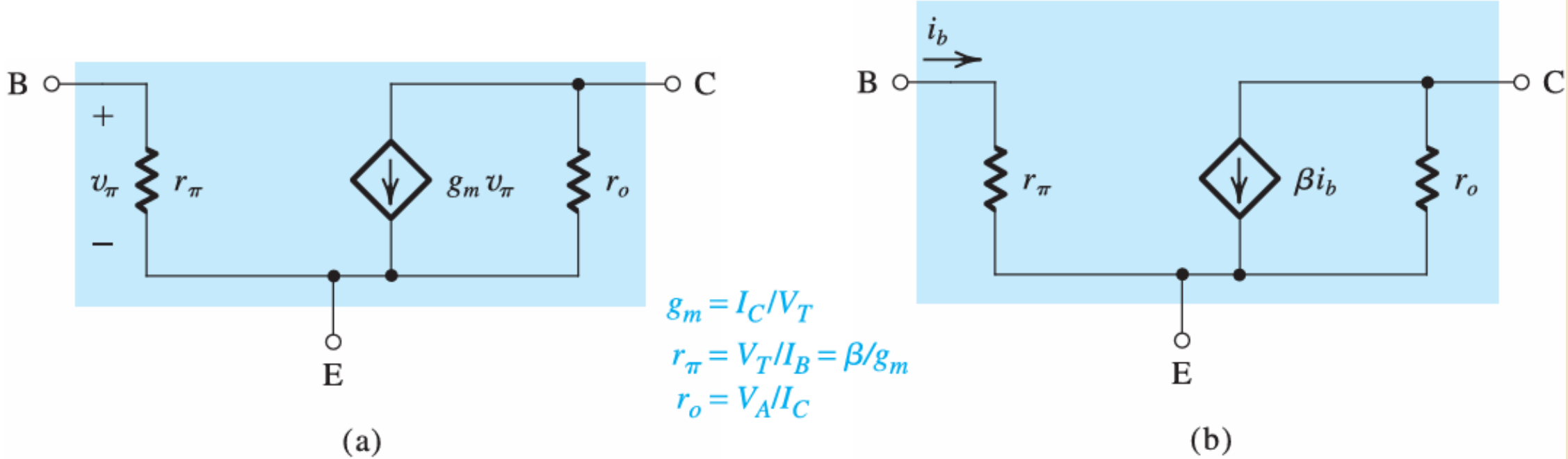
Two slightly different versions of the hybrid-π model for the small-signal operation of the BJT. The equivalent circuit in (a) represents the BJT as a voltage-controlled current source (a transconductance amplifier), and that in (b) represents the BJT as a current-controlled current source (a current amplifier).

$$i_e = \frac{v_{be}}{r_\pi} + g_m v_{be} = \frac{v_{be}}{r_\pi}(1 + g_m r_\pi)$$
$$= \frac{v_{be}}{r_\pi}(1 + \beta) = v_{be} \Big/ \left(\frac{r_\pi}{1+\beta}\right)$$
$$= v_{be}/r_e$$

A slightly different equivalent-circuit model can be obtained by expressing the current of the controlled source ($g_m v_{be}$) in terms of the base current i_b as follows:

$$g_m v_{be} = g_m(i_b r_\pi)$$
$$= (g_m r_\pi) i_b = \beta i_b$$

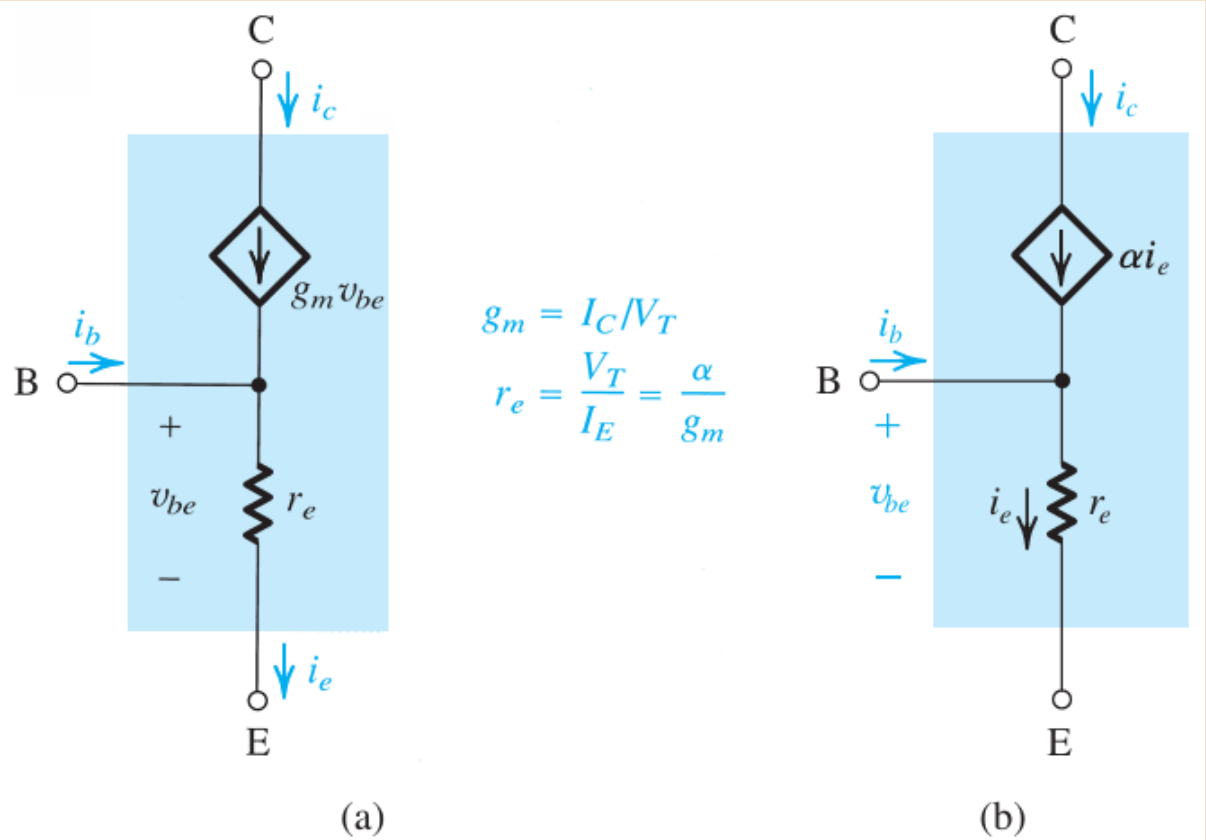
Considering the Early effect, and the resistance $r_o = V_A / I_C$ between collector and emitter, hybrid-π model for the small-signal operation of the BJT is shown below which is widely used for the analysis. Here v_{be} is renamed as v_π



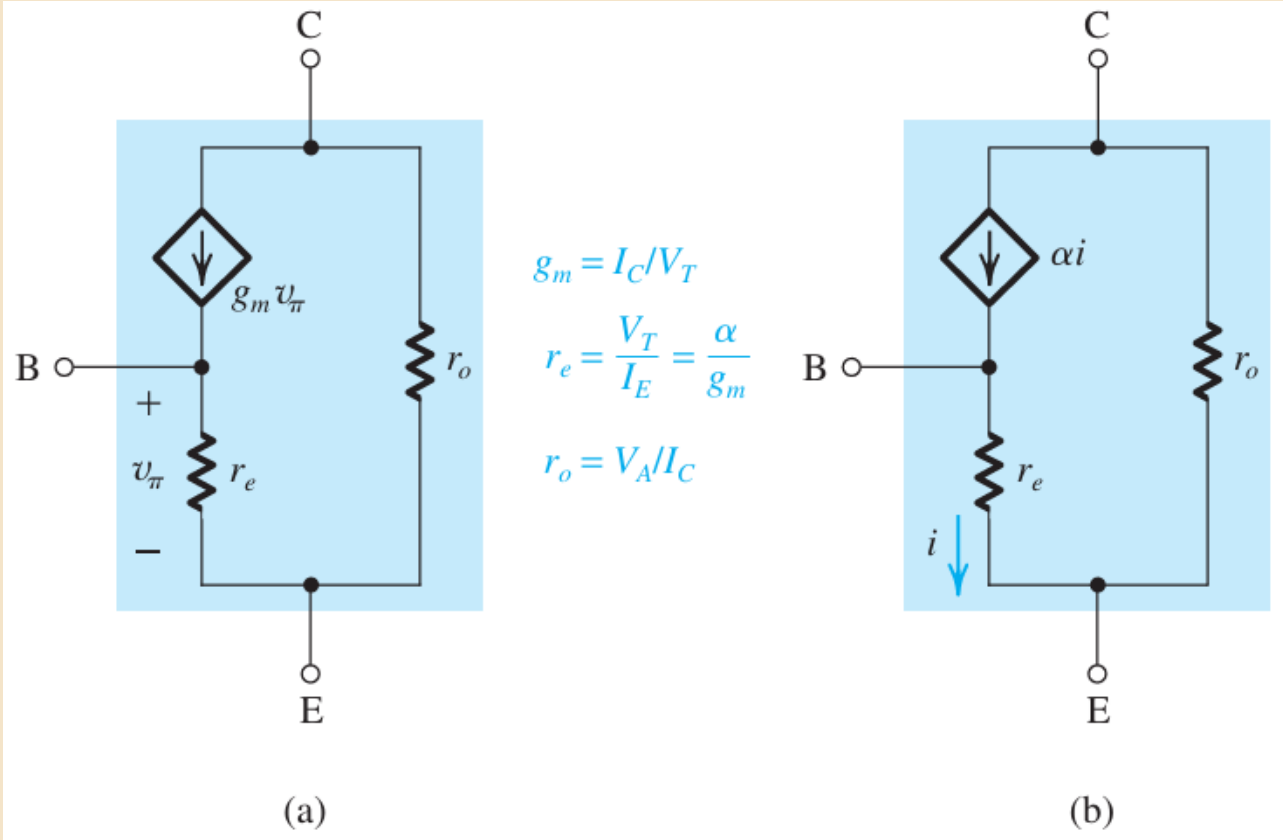
including r_o in the BJT model causes the voltage gain of the conceptual amplifier becomes

$$\frac{v_o}{v_{be}} = -g_m (R_C \parallel r_o)$$

Including r_o somewhat reduces the magnitude of the gain



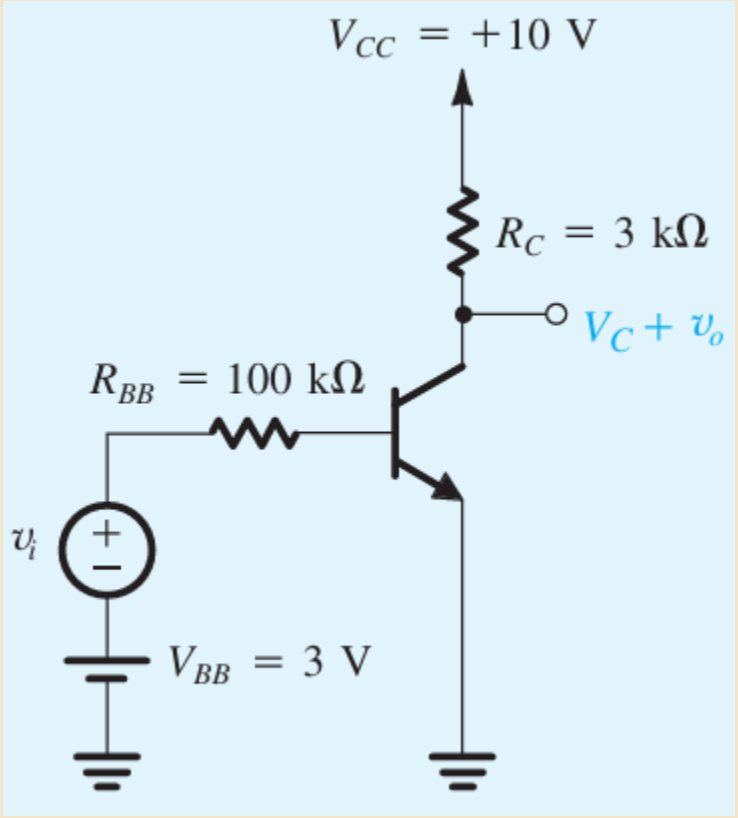
T Model of the BJT.
(a) Voltage-controlled current source representation
(b) Current-controlled current source representation



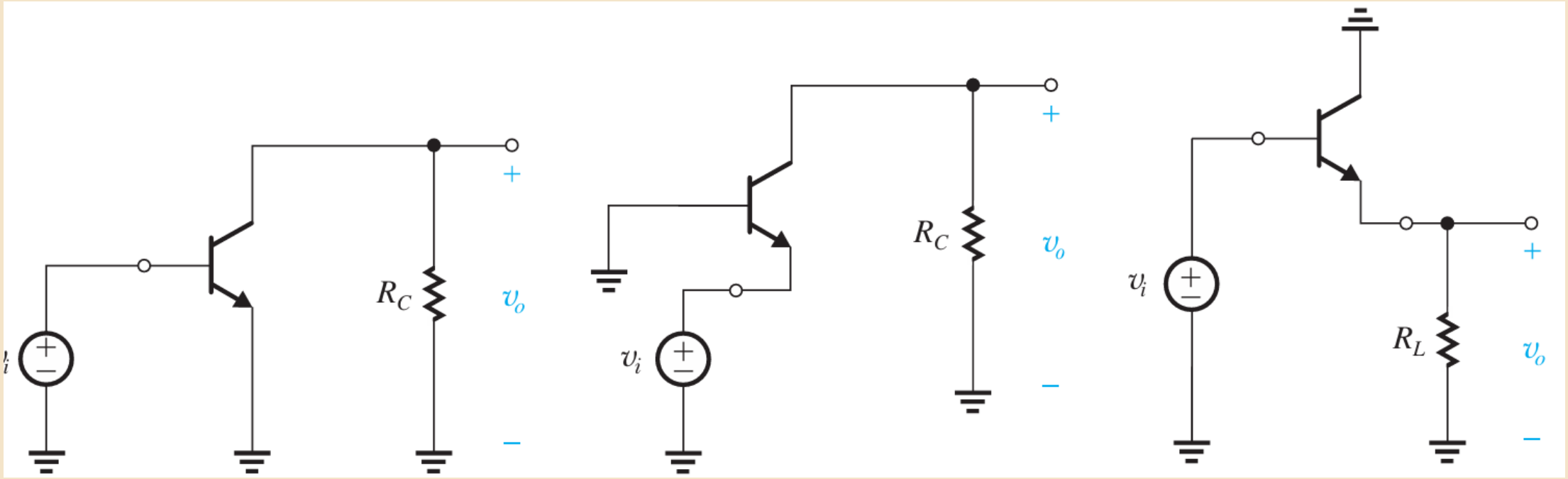
T Model of the BJT considering r_0

$$g_m v_{be} = g_m(i_e r_e)$$
$$= (g_m r_e) i_e = \alpha i_e$$

Analyze the transistor amplifier shown in Fig. to determine its voltage gain vo /vi . Assume β = 100 and neglect the Early effect.



Basic Configurations

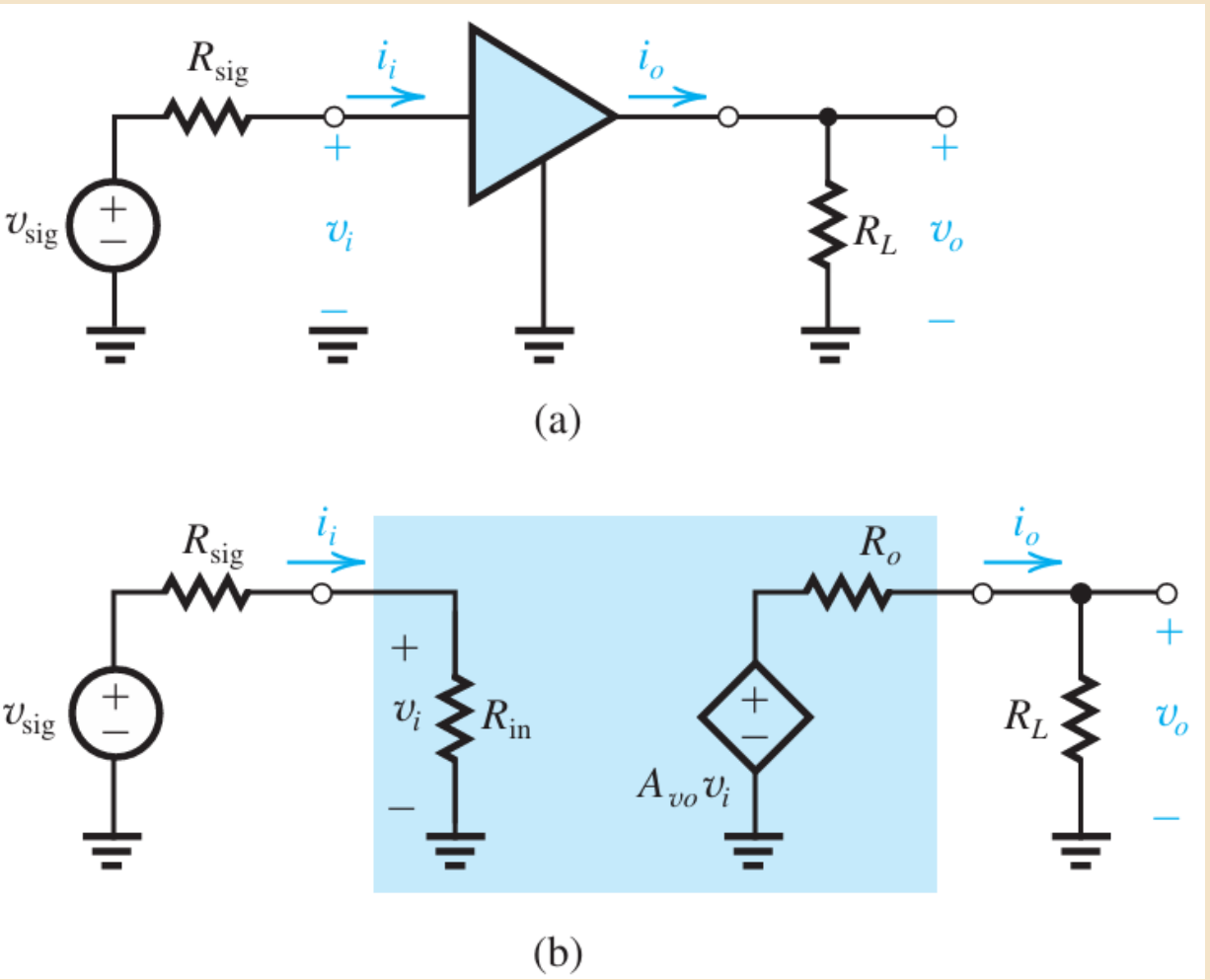


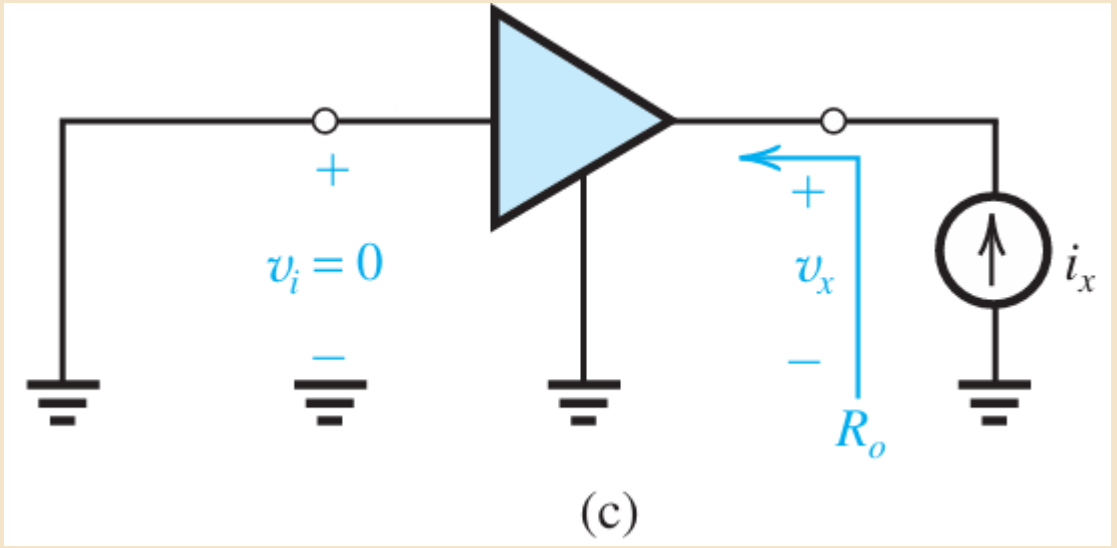
Common-Emitter (CE)

Common Base (CB)

Common Collector (CC)

Characterizing Amplifiers





Characterization of the amplifier as a functional block:

(a) Anamplifierfedwithavoltagesignal v_{sig} having a source resistance R_{sig}, and feeding a load resistance R_L

(b) equivalent-circuit representation of the circuit in (a)

(c) determining the amplifier output resistance R_o .

In Fig.b, the input resistance R_{in} represents the loading effect of the amplifier input on the signal source.

Parameters: characterizing amplifier performance: Input Resistance, Open circuit voltage gain and Output resistance

Input Resistance

R_{in} and R_{sig} forms a voltage divider that reduces v_{sig} to the value v_i that appears at the amplifier input,

$$R_{\text{in}} \equiv \frac{v_i}{i_i} \qquad v_i = \frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{sig}}} v_{\text{sig}}$$

Open-circuit voltage gain A_{vo}; defined as $A_{vo} \equiv \left.\frac{v_o}{v_i}\right|_{R_L=\infty}$

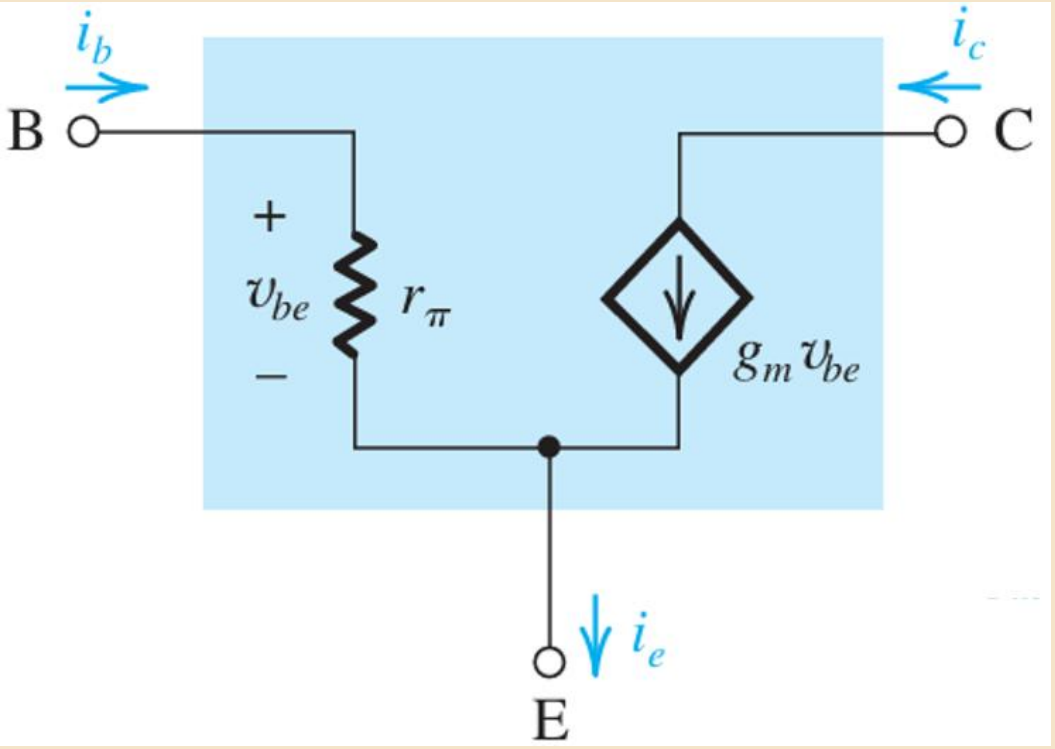
Output resistance R_o is the resistance seen looking back into the amplifier output terminal with v_i set to zero. From Fig. (C), $R_o = \frac{v_x}{i_x}$

Because R_o is determined with $v_i = 0$, the value of R_o does not depend on R_{sig} .

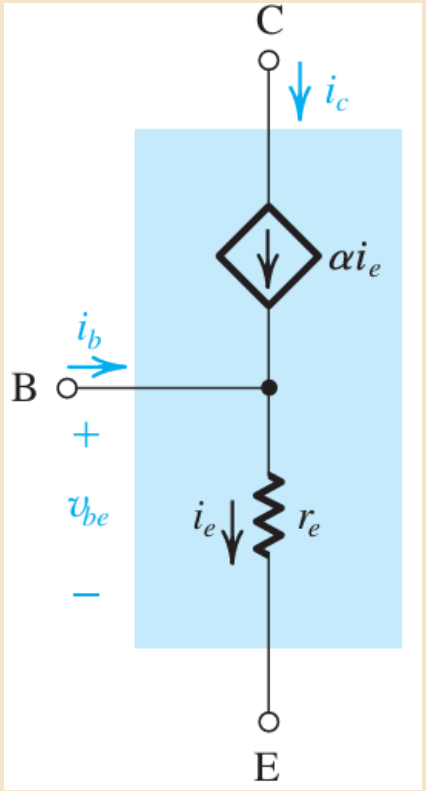
Output voltage v_o can be found from $v_o = \frac{R_L}{R_L + R_o} A_{vo} v_i$

Thus the voltage gain of the amplifier proper, A_v, can be found as $A_v \equiv \frac{v_o}{v_i} = A_{vo} \frac{R_L}{R_L + R_o}$

Overall voltage gain, G_v , $G_v \equiv \frac{v_o}{v_{\text{sig}}}$ $\qquad G_v = \frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{sig}}} A_{vo} \frac{R_L}{R_L + R_o}$

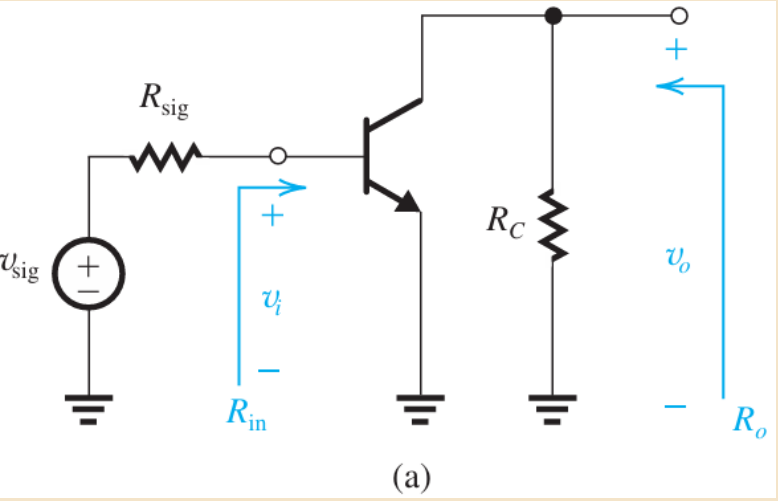


The hybrid-π small-signal model



T - Model

The Common Emitter Amplifier



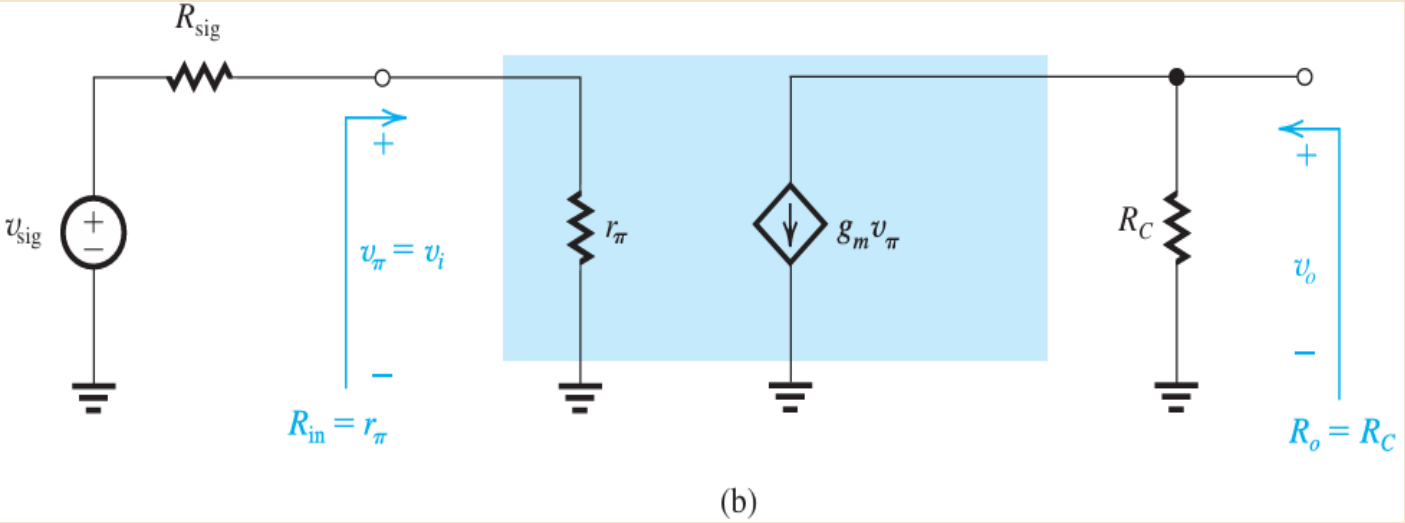
$$v_i = \frac{r_\pi}{r_\pi + R_{sig}} v_{sig}$$

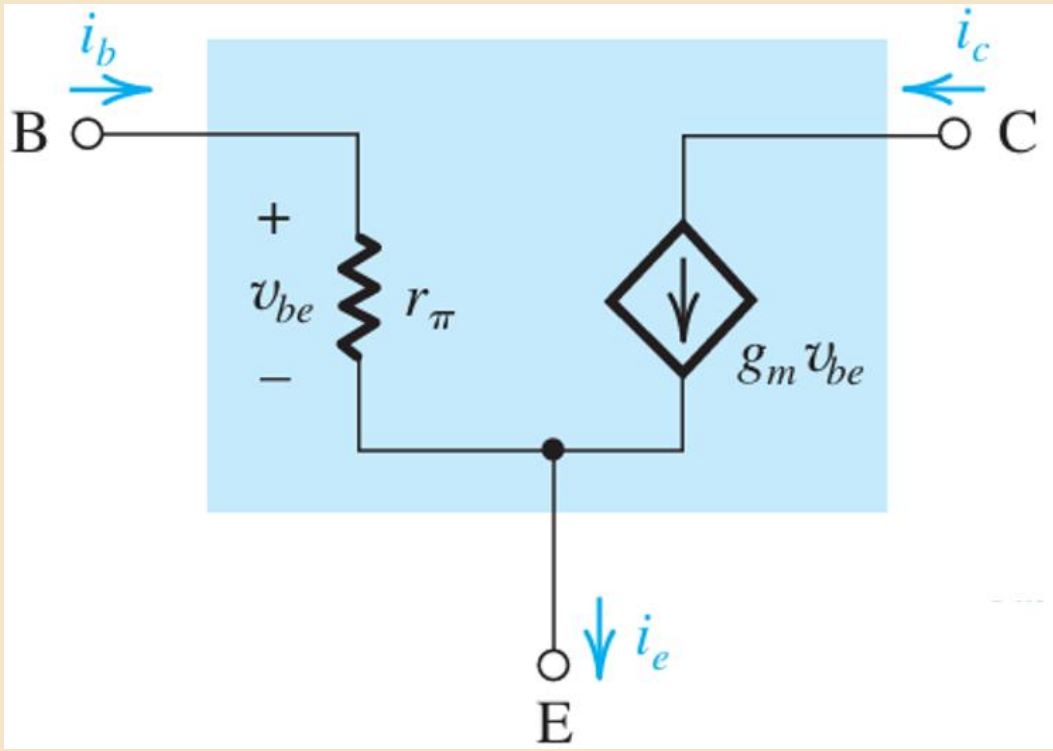
$$v_\pi = v_i$$

$$v_o = -g_m v_\pi R_C$$

$$A_{vo} \equiv \frac{v_o}{v_i} = -g_m R_C$$

$$R_o = R_C$$





With a load resistance R_L connected across R_C,

$$A_v = -g_m (R_C \| R_L)$$

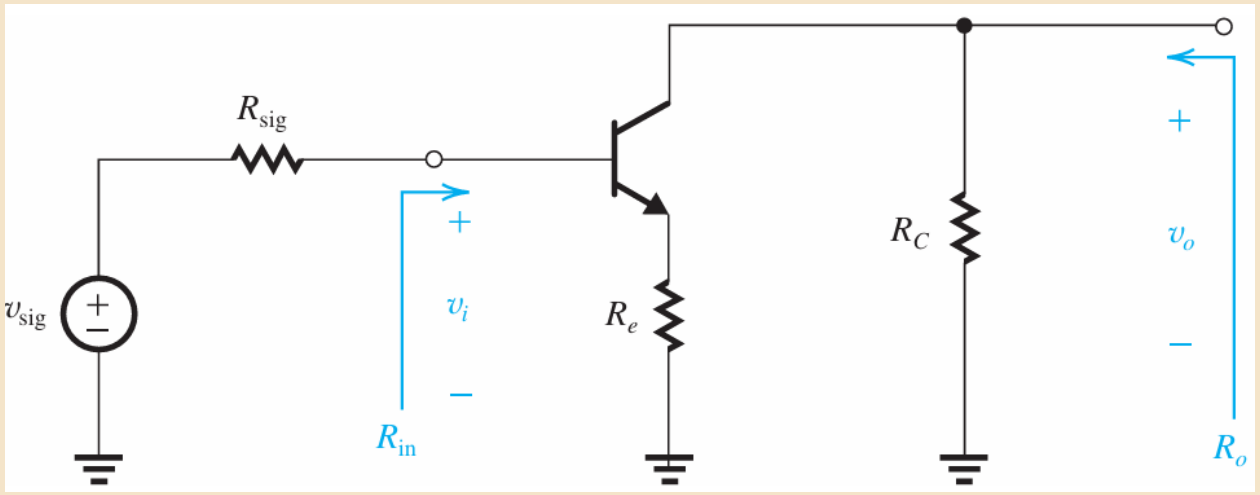
$$G_v \equiv \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i}$$

$$G_v = -\frac{r_\pi}{r_\pi + R_{sig}} g_m (R_C \| R_L)$$

(a) Common-emitter amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} . The bias circuit is omitted. (b) The common-emitter amplifier circuit with the BJT replaced by its hybrid-π model.

A CE amplifier utilizes a BJT with $\beta = 100$ is biased at $I_C = 1$ mA and has a collector resistance $R_C = 5\,\text{k}\Omega$. Find R_{in}, R_o, and A_{vo}. If the amplifier is fed with a signal source having a resistance of 5 kΩ, and a load resistance $R_L = 5\,\text{k}\Omega$ is connected to the output terminal, find the resulting A_v and G_v. If $\hat{v}_\pi$ is to be limited to 5 mV, what are the corresponding $\hat{v}_{\text{sig}}$ and $\hat{v}_o$ with the load connected?

CE Amplifier with Emitter resistance

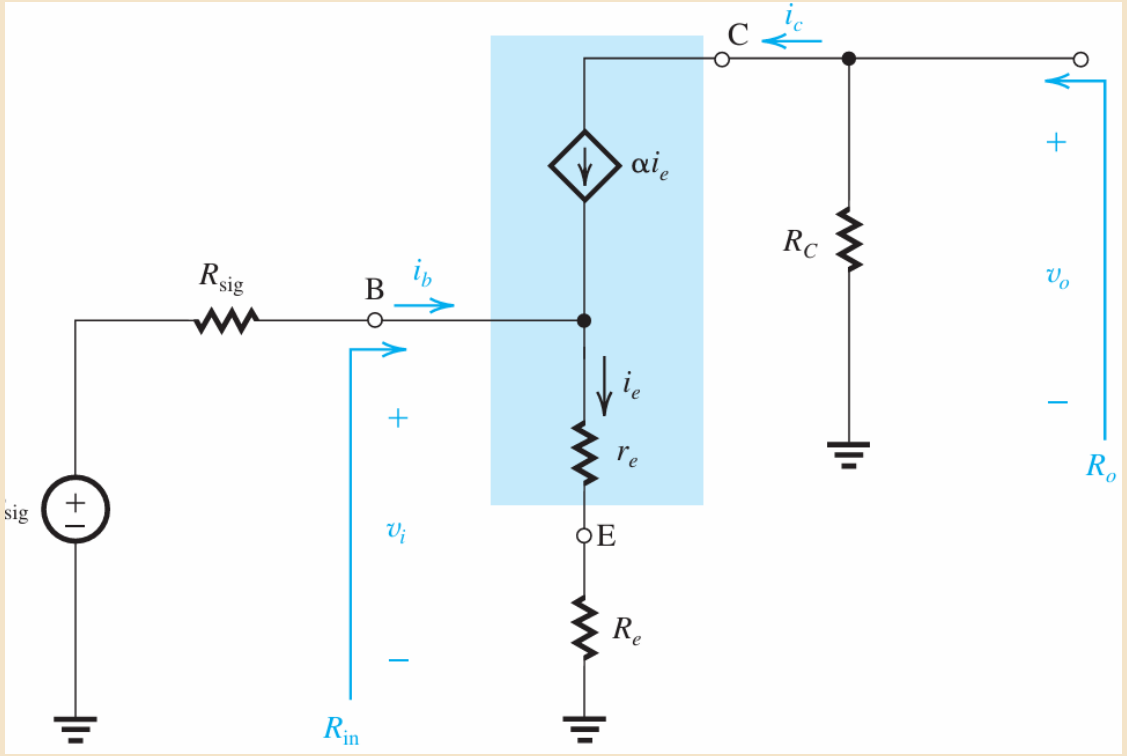


circuit without bias details

$$R_{in} \equiv \frac{v_i}{i_b} \qquad i_b = (1-\alpha)i_e = \frac{i_e}{\beta+1} \qquad i_e = \frac{v_i}{r_e + R_e}$$

$$R_{in} = (\beta+1)(r_e + R_e)$$

The input resistance looking into the base is (β +1) times the total resistance in the emitter, and is known as the **resistance-reflection rule**. The factor(β +1) arises because the base current is1/(β +1) times the emitter current.



Equivalent circuit with the BJT replaced with its T model

T model provides a simpler analysis and should be employed whenever there is a resistance in series with the emitter.

Including a resistance R_e in the emitter can substantially increase R_{in}

$$\frac{R_{\text{in}}(\text{with } R_e \text{ included})}{R_{\text{in}}(\text{without } R_e)} = \frac{(\beta+1)(r_e+R_e)}{(\beta+1)r_e} = 1+\frac{R_e}{r_e} \simeq 1+g_m R_e$$

The circuit designer can use the value of Re to control the value of R_{in}

$$v_o = -i_c R_C = -\alpha i_e R_C$$

$$A_{vo} = -\alpha\frac{R_C}{r_e+R_e} = -\frac{\alpha}{r_e}\frac{R_C}{1+R_e/r_e} = -\frac{g_m R_C}{1+R_e/r_e} \simeq -\frac{g_m R_C}{1+g_m R_e}$$

Thus, including Re reduces the voltage gain by the factor ($1+g_m R_e$), which is the same factor by which R_{in} is increased. This points out an interesting trade-off between gain and input resistance, a trade-off that the designer can exercise through the choice of an appropriate value for R_e.

$$R_o = R_C$$

$$A_v = A_{vo}\frac{R_L}{R_L+R_o} = -\alpha\frac{R_C}{r_e+R_e}\frac{R_L}{R_L+R_C} = -\alpha\frac{R_C \parallel R_L}{r_e+R_e}$$

$$G_v = \frac{R_{\text{in}}}{R_{\text{in}}+R_{\text{sig}}} \times -\alpha\frac{R_C \parallel R_L}{r_e+R_e}$$

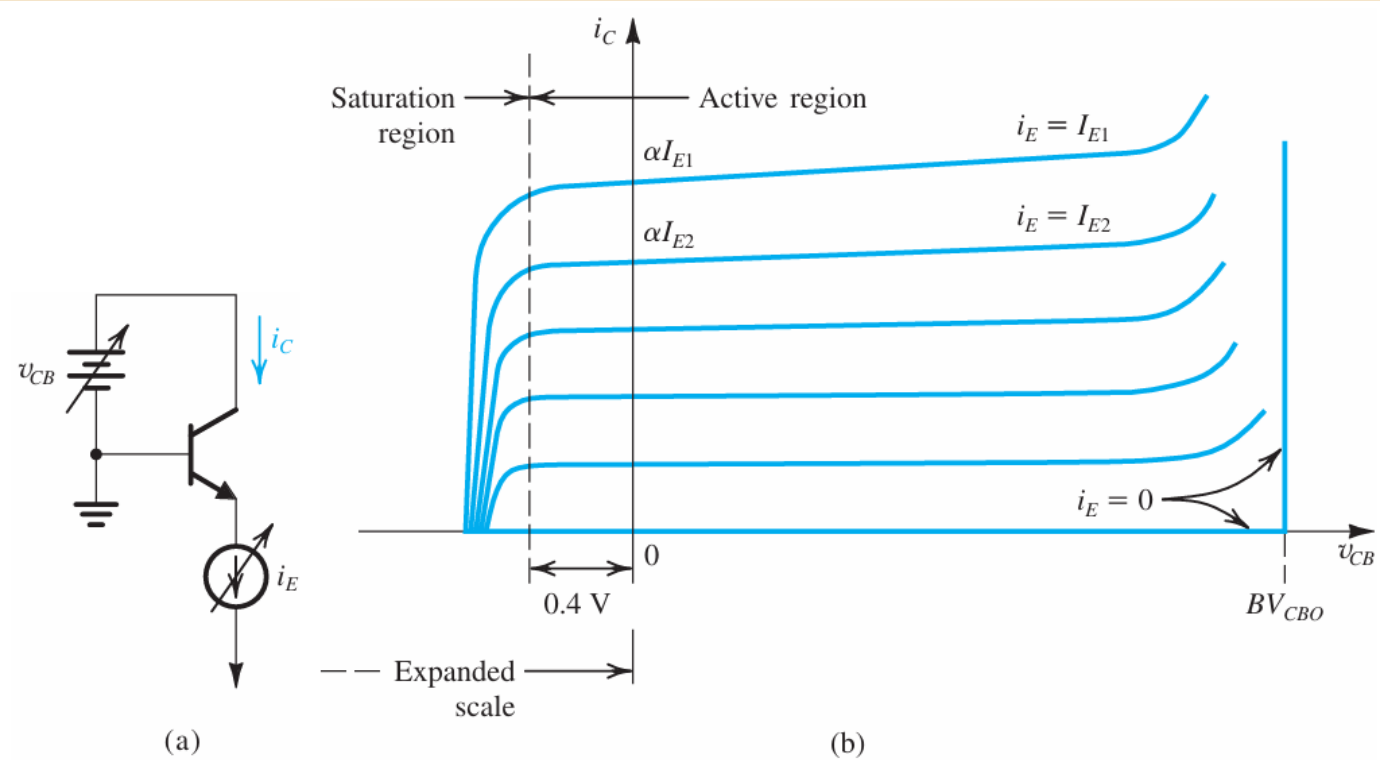
As $R_{\text{in}} = (\beta+1)(r_e+R_e)$

$$G_v = -\beta\frac{R_C \parallel R_L}{R_{\text{sig}}+(\beta+1)(r_e+R_e)}$$

To summarize, including a resistance Re in the emitter of the CE amplifier results in the following characteristics:

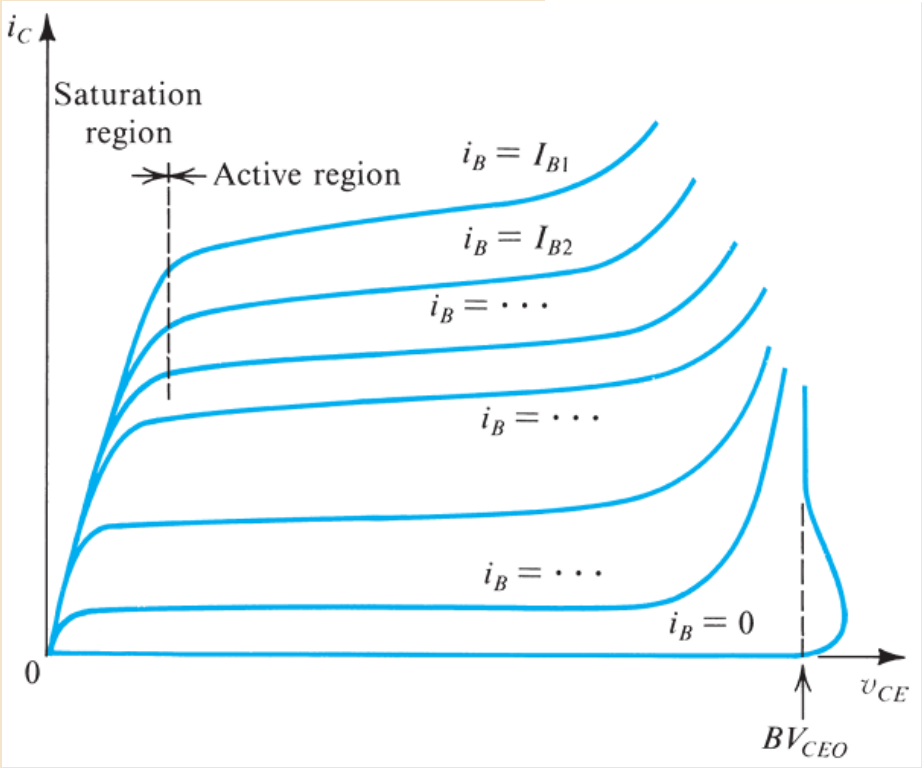
1. The input resistance Rin is increased by the factor $(1+g_m R_e)$.
2. The voltage gain from base to collector, Av , is reduced by the factor $(1+ g_m R_e)$.
3. For the same nonlinear distortion, the input signal vi can be increased by the factor $(1+ g_m R_e)$.
4. The overall voltage gain is less dependent on the value of β.
5. The high-frequency response is significantly improved

Transistor Breakdown and Temperature Effects



The BJT common-base characteristics including the transistor breakdown region

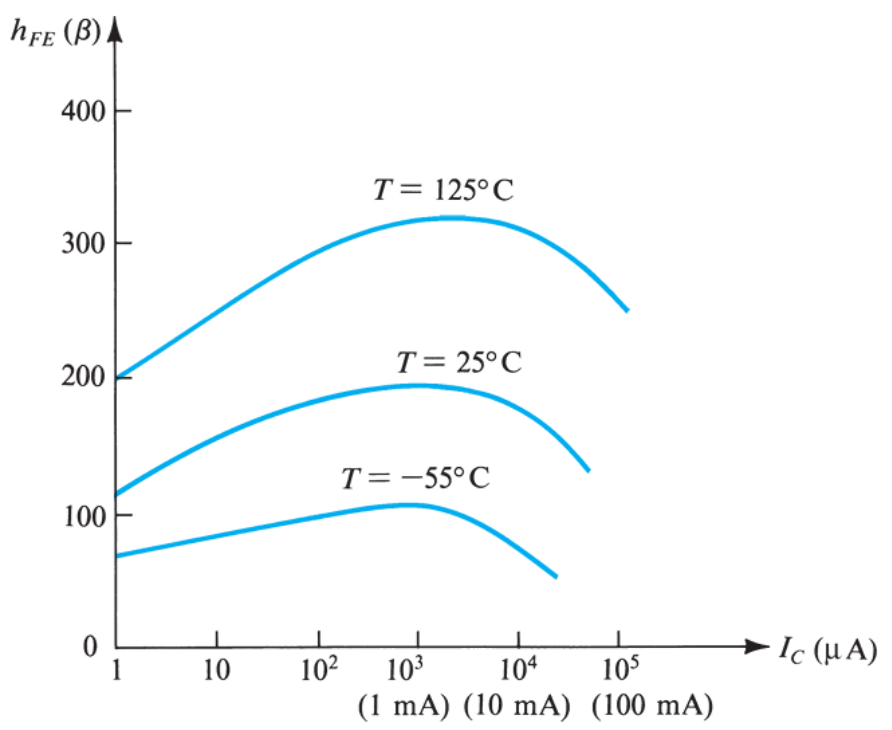
The i_C–v_{CB} characteristics in Fig. (b) indicate that for i_E =0 (i.e., with the emitter open-circuited) the collector–base junction breaks down at a voltage denoted by BV_{CBO}. For i_E >0, breakdown occurs at voltages smaller than BV_{CBO}. Typically, for discrete BJTs, B_{VCBO} is greater than 50 V.



The BJT common-emitter characteristics including the breakdown region

In common-emitter characteristics, breakdown occurring at a voltage BV_{CEO}. Here, although breakdown is still of the avalanche type, the effects on the characteristics are more complex than in the common-base configuration. BV_{CEO} is about half BV_{CBO}. On transistor data sheets, BV_{CEO} is sometimes referred to as the sustaining voltage LV_{CEO}.

Dependence of β on I_C and Temperature



Typical dependence of β on IC and on temperature in an integrated-circuit npn silicon transistor intended for operation around 1 mA.

At low I_C: β is relatively small. As I_C increases, β increases.

At moderate I_C: β reaches a maximum value.
This is the region where the transistor has its highest current gain.

At high I_C: β starts to decrease.
Therefore, operating the transistor at very high collector current reduces its current gain.
For example, at $25°C$, β reaches approximately 190–200 around $I_C \approx$ 1–2 mA, and then decreases as I_C becomes large.

Therefore, an increase in β causes I_C to increase.
The increased I_C can cause further heating, which can change the transistor currents again. This is one of the reasons proper bias stabilization and thermal stability are important in BJT circuits.

So, we should not assume that a BJT always has a constant β, such as 100. The actual β depends on the operating current, temperature, and individual transistor.

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